

CITATION: Butler v. Royal Victoria Hospital, 2017 ONSC 2792
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ONTARIO
SUPERIOR COURT OF JUSTICE

BETWEEN:

Sarah Butler, Luke Butler, James Butler and
Isaac Butler, minors by their Litigation
Guardian Jaye Butler and Robert Butler and
the said Jaye Butler

Plaintiffs

– and –

Royal Victoria Hospital, P. Bates, K. Rina,
K. Corvari, W. Dekleyne, A. Fenton, A.M.
Cichowicz, T. Haney, J. Stuart, E. Reyes, C.
Stewart, S. Causey, T. Whitty, V. Ho, and
M. Hazlett

Defendants

Hilik Y. Elmaliah, Jeremy Syrtash and
Michael Hershkop for the Plaintiffs

Valerie Wise and Julia Lauwers, for the
Defendants

**HEARD: May 16, 17, 18, 19, 20, 24, 25,
26, 27, 30, 31, June 1, 2, 3, 6, 7, 9, 10, 13,
14, 15, 16, September 7, 23 and October 7,
2016**

REASONS FOR DECISION

J.R. McCARTHY J.:

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Background

- [1] The minor Plaintiff Sarah Butler was born January 26, 2007. She is the daughter of Mrs. Jaye Butler and Mr. Robert Butler. The couple has three other children: James, Isaac and Luke. Luke is Sarah's twin.
- [2] Mrs. Butler became pregnant with the twins in May 2006. Prior to her attendance at Royal Victoria Hospital (RVH) early in the morning of January 26, 2007, Mrs. Butler's pregnancy was normal.
- [3] Within 30 minutes of her arrival at RVH, Twin A (Luke) was delivered vaginally. At around 1am, the two nurse Defendants proceeded to artificially rupture the membranes of Twin B (Sarah). This resulted in a compression of the umbilical cord for some 25 minutes. Sarah was delivered by emergency Caesarean section (C-section) at 1:29am.
- [4] Sarah was depressed at birth; she required resuscitation with intubation, positive pressure ventilation and Narcan. Her Apgar scores were low. Sarah's initial arterial cord blood gas was abnormal with a pH of 6.64. Sarah displayed no spontaneous movements or efforts at breathing until more than 5 minutes following delivery. She emitted her first gasp 5 minutes following delivery; her respirations began approximately 10 minutes later. At about 26 minutes following delivery, seizure-like activity was noted. Sarah did not move her limbs spontaneously until about 2½-3 hours following delivery. Sarah remained intubated for 10 hours; she was then self-extubated and weaned to room air.
- [5] An electroencephalogram (EEG) conducted on January 29, 2007 revealed a generalized encephalopathy. Sarah remained at RVH until February 15, 2007. The discharge diagnosis stated that Sarah experienced perinatal asphyxia and hypoxic ischemic encephalopathy (HIE) II with seizures. Sarah was referred to the RVH Neonatal Follow-Up Clinic and to Dr. William Logan, a Paediatric Neurologist at the Hospital for Sick Children (HSC) Neurology Clinic. Dr. Logan first saw Sarah in July 2007. Over a number of visits, he rated Sarah's perinatal asphyxia as "very severe", and also diagnosed developmental delay, cognitive delay and ataxia. One of Sarah's other treating physicians, Dr. Rania Hiram, noted residual neurological deficits in August 2007 and developmental delay in February 2008.
- [6] Dr. Logan ordered a Magnetic Resonance Imaging (MRI) scan in July 2008. The MRI scan showed increased signal in the white matter and prominence of the temporal horns of the lateral ventricles with under rotation of the hippocampi. The reviewing radiologist, Dr. Manohar Shroff, considered these findings nonspecific and of unknown significance. Since the MRI scan did not show the damage expected of perinatal asphyxia, evaluation for other genetic and metabolic causes was undertaken. These evaluations showed no abnormalities.
- [7] In September 2008, Dr. Hiram diagnosed Sarah with hypotonic cerebral palsy.
- [8] A second MRI was arranged by Plaintiffs' counsel and conducted on May 31, 2012. Dr. Alan Hill, an expert in Paediatric Neurology, found this MRI to reveal that there had been no improvement in the white matter of the brain. Since a great deal of myelination would have taken place in a normal five-year-old child, he attributed this result to a pathological process.

Dr. Robert Zimmerman, an expert in Paediatric Neuroradiology, concurred that these results suggested that Sarah's injury would be a lifelong one.

- [9] Dr. Emily Tam started treating Sarah in 2014. Dr. Tam considered the clinical history, the 2008 MRI and the normal genetic testing. She noted that brain imaging is not always completely "classic" for perinatal asphyxia. In the absence of developmental regression or any signs of other etiologies, Dr. Tam found that Sarah's symptoms were likely caused by the hypoxic ischemic event at birth.

The Issues

- [10] The Defendant RVH (hereinafter referred to as the Defendant) concedes a breach of the standard of care by its hospital nurses in artificially rupturing the membranes of Twin B (Sarah) on the date in question. The remaining issues for determination by this court are causation and damages.

The Plaintiffs' Position on the Issues

- [11] The Plaintiffs contend that, as a direct result of the rupture of the membranes, Sarah suffered HIE, a type of brain injury which has resulted in cerebral palsy, cognitive impairments and behavioural problems. Sarah's intrapartum hypoxic-ischemic brain injury is the etiological cause of all of her impairments and deficits.
- [12] These impairments and deficits affect gross and fine motor skills, speech, cognition, learning and behaviour. The deficits and impairments are serious and permanent. Sarah is entitled to recover general non-pecuniary damages for her pain and suffering.
- [13] Sarah is competitively unemployable and is entitled to damages for future loss of income. In order to cope with her on-going disabilities and to enhance her quality of life, she will require physiotherapy, speech-language therapy, occupational therapy and other goods and services for the balance of her lifetime.
- [14] Finally, Sarah will never be able to live or function independently, leaving her in need of 24-hour care for the remainder of her life. Her family members are entitled to be compensated, not just for loss of care, guidance and companionship but for the extraordinary care and supervision that Sarah has required beyond what she would have needed had she not sustained the injuries in question.

The Defendant's Position on the Issues

- [15] The Defendant admits that the hospital nurses ought not to have ruptured the membranes on the date in question. The Defendant does not dispute that Sarah suffered a neonatal asphyxia that caused brain injury; nor does it dispute that the neonatal presentation qualified as HIE. The Defendant concedes that the artificial rupturing of the membranes caused Sarah's ataxic hypotonic cerebral palsy.

[16] The Defendant argues that the fact that Sarah suffered asphyxia or HIE in the neonatal period does not establish or prove that this event is responsible for all of Sarah's current issues. The Plaintiffs have not proven on a balance of probabilities that Sarah would have had no attentional, cognitive and behavioural issues "but for" the event of neonatal asphyxia. Inherent in the "but for" test is the requirement to show that the Defendant's negligence was necessary to bring about the injury – in other words, that the injury would not have occurred without the Defendant's negligence. Sarah's cognitive and behavioural deficits would have emerged, without the Defendant's negligence, because of genetically inherited Attention Deficit Hyperactive Disorder (ADHD).

[17] The causation issue has been complicated by the Plaintiffs' unwillingness to produce important documentation, including the radiology report from the 2012 MRI, and certain school records of Sarah and her siblings. The causation issue is further complicated by the Plaintiffs':

- Refusal to have Sarah participate in behavioural therapy;
- Refusal to have further genetic testing done of Sarah and members of her family;
- Failure to follow a proper course in the administration or titration of ADHD medication;
- Decision to remove Sarah from the care of her treating physicians since November 2014;
- Refusal to have Sarah undergo recommended psycho-educational and neuropsychological assessments; and
- Provision of misleading or inaccurate information to Sarah's treating doctors.

[18] On the issue of damages, while holding to its position that the Plaintiffs have failed to meet the burden of proving that Sarah would not have had cognitive delays and other issues but for the asphyxia, the Defendant has suggested a range of damages that would fairly compensate Sarah for whatever injuries were caused by the nurses' negligence. Those injuries, however, would be limited to asphyxia, brain injury, HIE, cerebral palsy and borderline microcephaly.

[19] Even if all of her present problems are the responsibility of the Defendant, there is no real and substantial possibility that Sarah will require the host of treatment and services suggested by the Plaintiffs. In assessing damages, the court should bear in mind the significant role being played by the ADHD in Sarah's presentation. The court must accept the evidence that proper treatment with medication has an 80% chance of addressing Sarah's ADHD issues. The court must take into account this failure to mitigate and also assess Sarah's prospects going forward if proper treatment is afforded to her.

[20] The second prong of the ADHD analysis must be a contingency reduction on damages for the likelihood that both Sarah's ADHD and her cognitive delay would have occurred despite birth injuries. An appropriate reduction for this contingency would be at least 24%.

[21] Finally, the Defendant argues that the Plaintiffs have greatly exaggerated Sarah's present-day needs and limitations and have failed to demonstrate that her present-day condition will endure throughout her lifetime.

The Plaintiffs' Evidence on Causation

Jaye Butler

[22] Ms. Butler denied any family history of genetic disorder. After a normal pregnancy, Ms. Butler gave birth to Luke and Sarah on January 26, 2007. Luke's birth was normal. Sarah's birth involved the complications set out above. Ms. Butler provided a summary of Sarah's current deficits and impairments which included the following:

- Right hand tremors;
- Inability to pour a drink;
- Illegible hand printing;
- Clumsiness;
- Coordination deficits and a tendency to fall;
- Poor fine motor skills (e.g. unable to manipulate buttons, shoe laces, zippers, and seat belts);
- Dependence on a booster seat;
- Difficulty negotiating stairs;
- Drooling;
- Unclear speech;
- Difficulties with self-grooming and dressing;
- Bowel movements in underwear;
- Requires assistance with some aspects of toileting;
- Awakens two-three times per night;
- Requires assistance cutting food;
- Requires supervision while eating;
- Has no appreciation of danger;
- Cannot swim;
- No concept of time or differences in the value of currency;
- Temper tantrums;
- Does not know home address or days of the week;
- No real friends;
- Receives no invitations for play dates, sleepovers, or birthday parties;
- No concept of etiquette or social boundaries;
- Resistant to change in routine;
- Unable to carry out tasks or chores;
- Concentration and behavioural issues at school;
- Subject to bullying and teasing at school; and
- Requires constant supervision.

- [23] Ms. Butler came to understand that Sarah had developed hypotonic cerebral palsy as a result of the lack of oxygen flowing to her brain at birth. No treating physician has ever suggested that Sarah had ADHD.
- [24] With regards to her other children, Ms. Butler advised that James does well at school; he is socially and recreationally active.
- [25] Isaac has behavioural issues. He is rebellious. Isaac is nevertheless quite popular and always has a girlfriend. Although Isaac was diagnosed with ADHD in 2009, that diagnosis was sought to grant him access to a computer at school. Isaac is a hunter and fisherman. He has passed a hunting course and is a licensed scuba diver. Isaac has moved from an Individual Education Plan (IEP) to regular classes in high school.
- [26] Luke is doing well. He attends birthday parties, sleepovers and has friends. He is active in sports. His performance at school is adequate. He is attentive, organized and focused. Luke is pulling ahead of his twin, Sarah, both academically and socially.
- [27] In cross-examination, Ms. Butler agreed that there was no hand railing for the main floor stairs in their home in August 2013. As well, Ms. Butler was shown clips of video surveillance taken of her family. In the May 29, 2015 clip, Sarah is captured doing a little dance and attempting a cartwheel. She is twirling. Sarah entered the school van on her own. In a clip from her school yard, Sarah is seen running and doing a twirl, interacting with other girls and lifting up a little boy. In the June 4, 2015 clip, Sarah can be seen bending over to help pull weeds in the family's front yard. Later, she is seen stepping into a van.
- [28] Ms. Butler conceded that both Sarah and Luke were in diapers until age two; both entered pre-school in 2009. All occupational therapy, speech-language therapy and physiotherapy services were provided to Sarah in that setting. Sarah entered into junior kindergarten (JK) in 2011; it was all day, every day. Sarah did not sleep through the night in these years: she would awake thirsty, with nightmares or with wet diapers. During growth spurts in the JK and senior kindergarten (SK) years, Sarah would have a bowel accident every one to two weeks.
- [29] Sarah requires supervision while bathing. Sarah does not independently negotiate the hallway to the bathroom in the dark. Sarah stopped wearing an orthotic by JK. Sarah can manage ½ inch buttons and zippers using a special pull string. Sarah's daily routine after school consists of: a 4pm dinner, some play time, a bath, a snack and bed by 7:30pm. Sarah is not involved in extra-curricular activities.
- [30] Dr. Logan advised Ms. Butler that the pattern seen on the 2008 MRI of the brain was not consistent with perinatal asphyxiation and that the etiology for Sarah's profile was uncertain.
- [31] In 2011, Ms. Butler advised Dr. Logan of Sarah's concentration and attention problems as well as the improvement in her fine motor skills. She did not mention the bowel incontinence issue to Dr. Logan.

[32] Ms. Butler was asked by her lawyer not to mention the 2012 MRI to either Dr. Tam or Dr. Logan.

[33] Ms. Butler advised Dr. Tam that Sarah's siblings were developmentally normal.

[34] Dr. Hiram recommended an EEG, a consultation with an ear, nose and throat specialist and a consultation with a neuropsychologist. The family did not follow up on these recommendations. Sarah last saw Dr. Hiram in 2011. Since the commencement of this litigation, medical consultations have been limited to Dr. Trevor Poole and neurologists at HSC.

[35] The description of Sarah in the psychological assessment report dated May 17, 2011 was accurate. Sarah struggled with attention and focusing; she was impulsive and distracted; she was bossy and required prompting.

[36] The features describing Sarah in the pre-school Individual Family Service Plan Review of May 2011 were also accurate.

[37] Sarah's Grade One report cards expressed concerns with distracting behaviour and listening problems. The Strengths and Needs Committee (SNC) records indicated that Sarah was easily distracted and engaged in attention-seeking behaviour.

[38] This excerpt from her Grade Two report card was accurate: "Reminders are often required for Sarah to follow classroom expectations and routines. Her behaviour is beginning to interfere with her learning as well as the learning of her peers."

[39] In 2009, while he was in Grade 4, Isaac was diagnosed with a learning disability and ADHD. Isaac has experienced some problems with reading, writing and math; he has no issues with initiation, working memory, impulsivity, inattention and distractibility. While Isaac was placed in special education until Grade Eight, he has since graduated and moved on to an alternative school in Bradford. He has completed Grade Ten taking applied level courses. In 2014, Isaac was suspended one school day for being in a physical altercation with another student.

[40] Ms. Butler could not recall receiving the psychological assessment report of January 27, 2011 in which Luke was found to be at risk for developing a learning disability. Nor could she recall preparing a parent report for a speech and language assessment in June 2011. Luke did poorly in Grade Two. Luke progressed with difficulty in language, math and science in Grade Three.

[41] Ms. Butler expressed surprise that Luke had been identified with fine motor and language skills issues and that he had been flagged for possible special needs in June 2011. She could not recall the SNC meetings in SK, nor any of the concerns expressed by the participants. Ms. Butler agreed to a limited IEP for Luke's reading and written expression; however, she could not recall declining a language assessment in late 2015. Mrs. Butler was not aware that Luke was in a special education program. She contended that Luke is presently doing well.

- [42] In Sarah's case, there were documented concerns in JK with respect to her toileting, fine motor skills, speech, focus, attention and sensory needs. Concerns about choking and flight risk led to the development of a safety plan. Sarah was defiant with adult requests and required constant redirection.
- [43] Ms. Butler conceded that there is no mention of Sarah's incontinence issues either in the SNC reports or in any IEP.
- [44] Ms. Butler maintained that Sarah requires constant supervision at school. She advised Dr. Shevell that Sarah was impulsive and disruptive in a group. Ms. Butler denied talking to Dr. Shevell about any trial of medication. She could recall no discussion with any of the Defence experts about medication.
- [45] Sarah's trial of medication was prescribed by Dr. Poole. Ms. Butler administered the medication to Sarah one time daily as prescribed. Ms. Butler observed no change in Sarah. She reported this to Dr. Sharon Marcovitch in January 2016. The course of Ritalin began on May 10, 2016.
- [46] Ms. Butler was unaware that the Plaintiffs' future care cost expert proposes round the clock care for Sarah. She wishes for Sarah to be independent and would be willing to try treatments and medication if recommended by a pediatrician.
- [47] In re-examination, Mrs. Butler recalled that no treating physician ever recommended a further MRI scan. Dr. Tam advised that the inadequacy of oxygen at birth was the reason for Sarah's present symptoms.
- [48] In 2009, at the time of the ADHD assessment, Isaac was independent, sociable, even-tempered, creative; he liked sports, camping, building, the outdoors and physical education. Isaac's Grade Three report cards records satisfactory or good progress in independent work, initiative, cooperation with others and conflict resolution. On his Grade Four report card, Isaac received ratings of good or excellent in all learning skills areas. This was far different from Sarah's performance at that same stage.
- [49] Though I found Ms. Butler to be a credible witness, her testimony was not without problems. She was anxious to minimize the learning and behavioural issues of both Luke and Isaac. I find it troubling that a parent would willingly seek out or allow a diagnosis of ADHD for her child just to obtain a computer. She struck me as prone to exaggeration about some of Sarah's problems. I find it highly unlikely that she would not know that her own future care expert is advocating 24-hour care for Sarah when that represents a significant component of the claim being advanced by the Butler family.
- [50] Nevertheless, Ms. Butler obviously and understandably has the best interests of her family at heart; she was a good historian; she was not disagreeable or hostile under cross-examination. She cannot be blamed for what happened to her daughter at birth. She cannot be expected to agree with a diagnosis of ADHD when such a thing was never suggested by any of the treatment providers or school authorities. I accept her evidence as generally reliable and entitled to weight.

Robert Butler

[51] Mr. Butler was an excellent witness. He was candid, frank and forthright. He had an impressive recollection of the unfortunate events surrounding Sarah's birth and the family's subsequent interaction with RVH. He confirmed much of the description of Sarah's profile offered by his spouse. He understood from Dr. Tam that an MRI scan of the brain often would not show evidence of perinatal asphyxia.

[52] Mr. Butler maintained that Isaac was developmentally normal and did not have dyslexia.

[53] I accept that the family did not follow up on the referral to a neuropsychologist because of cost, wait time and the number of tests involved.

Dr. Max Perlman

[54] Dr. Perlman was called by the Plaintiffs as an expert in Neonatology. Dr. Perlman has a special interest in HIE and brain injury.

[55] Dr. Perlman opined that Sarah had suffered HIE of the acute total or near total type at birth. The compression of the umbilical cord following the rupture of the membranes prevented the perfusion of oxygenated blood for 25 minutes. This resulted in extreme intra-partum asphyxia.

[56] The diagnosis of HIE was supported by the clinical features present at or shortly after birth: severe depression requiring resuscitation, delayed onset in breathing, abnormal brain signals, multiple organ/system involvement, anemia and hypovolemia. Sarah's umbilical cord blood gas measurements were indicative of an extremely severe asphyxia, almost incompatible with life. These blood gas measurements were consistent with a 25-minute period of total hypoxia.

[57] With chilling precision, Dr. Perlman explained how the insult to the brain likely began immediately after the rupture of the membranes; whereas the injury to the brain began no earlier than seven minutes before birth and likely only three to four minutes before birth.

[58] Dr. Perlman ruled out other neonatal causes of the HIE. Typical outcomes from HIE include cerebral palsy, severe behavioural, cognitive and developmental impairments and microcephaly (a small head). Sarah's head circumference at 19 months was within the 2nd percentile. Sarah's profile fit perfectly with a history of HIE. Given the severe HIE in this case, one could have predicted with a high degree of certainty the outcome that followed.

[59] Dr. Perlman's opinion evidence was not eroded on cross-examination. Although Dr. Perlman did not meet with Sarah or her parents, I find that his review of the medical records was thorough and his interpretation of them was reliable. He maintained his position that developmental and behavioural problems were on the continuum of outcomes for HIE.

[60] Dr. Perlman agreed that in moderate to severe asphyxia insults to the brain, the typical site of brain injury would be basal ganglia and thalami; in the case of severe injury, the site of

injury would include the cerebral cortex. Dr. Perlman found no evidence in the records of any injury to the basal ganglia.

[61] Neither ataxic cerebral palsy nor attention deficit issues appear on the list of possible disabilities enumerated in a 2006 article co-authored by Dr. Perlman and Dr. Charlene Robertson entitled “Follow-up of the term infant after hypoxic-ischemic encephalopathy”. In fact, the article suggests that ataxic cerebral palsy is uncommonly attributable to HIE.

[62] Nevertheless, Dr. Perlman went on to explain how by age eight, most victims of HIE had significantly lower verbal and performance intelligence quotients (IQs), as well as delays in academics and motor domains.

[63] In addition, 71% of a subject study of HIE victims had cognitive and executive dysfunction. Dr. Perlman added that those studied would have been significantly less damaged than Sarah. Nevertheless, most of them were unemployed or unemployable and had serious problems with their daily life situation.

Dr. Alan Hill

[64] Dr. Hill was called by the Plaintiffs as an expert in Paediatric Neurology with a focus on Neonatal Neurology. He has researched and published extensively on the subject of HIE and its outcome.

[65] Dr. Hill confirmed that the artificial rupture of the membranes in this case resulted in compression of the umbilical cord. This manifested itself in a severe fetal bradycardia (slow heart rate) until Sarah was delivered by emergency C-section 25 minutes later. During this time, Sarah’s brain suffered a hypoxic-ischemic insult: “hypoxia” meaning not enough oxygen and “ischemia” meaning inadequate delivery of blood to the brain.

[66] As a result, Sarah was born in a severely compromised condition. The measurement of blood in the umbilical cord revealed a severe metabolic acidosis, indicative of an attempt by the fetus to produce the oxygen it was lacking; it was also consistent with a deprivation of oxygen lasting 25 minutes.

[67] Sarah required vigorous resuscitation; produced low Apgar scores; had seizure activity; displayed abnormal tone; and needing assistance with feeding. The child was thus severely compromised.

[68] The hypoxic-ischemic insult resulted in “encephalopathy” or abnormal function of the brain. Sarah therefore qualified for a diagnosis of HIE or neonatal encephalopathy. But for this hypoxic ischemic event, Sarah would have been normal – the fetus was otherwise healthy and well grown at birth.

[69] The consequences of the injury to the brain included the onset of ataxic cerebral palsy and cognitive/behavioural problems. The hypoxic ischemic insult in question was severe and near total. This pattern of injury to the brain involved certain structures. That pattern is not the same in every case but usually includes: the deep central grey matter in the mid-brain, the

thalamus and basal ganglia, the hippocampus, the vermis of the cerebellum and the white matter leading to the Rolandic cortex.

[70] The parts of the brain injured in Sarah's case are often associated with the subsequent development of motor impairments, cognitive/behavioural and learning problems, and with ADHD symptoms.

[71] Dr. Hill concluded that Sarah's deficits and abnormalities were caused directly by the hypoxic-ischemic brain injury.

[72] Dr. Hill's conclusions and opinions were comprehensively challenged in cross-examination. He believed that the 2008 MRI scan showed an abnormal signal in the white matter of the brain. This may have indicated incomplete myelination; however, by 2012, the unchanged signal seen in the white matter was more than likely related to a pathological process rather than to a normal variant of myelination.

[73] Dr. Hill expressed surprise that Dr. Shroff would have considered the prominence of the temporal horn and the under-rotation of the hippocampi, imaged on the 2008 MRI, to be of unknown significance. Dr. Hill disagreed that these were nonspecific findings. The investigation into other possible etiologies (including a genetic component) by the treating doctors at HSC was understandable given that the imaging study had failed to provide a definitive answer. The uncertainty surrounding the etiology arose from Dr. Shroff's interpretative report; Dr. Hill could not say whether any of these other treating doctors even looked at the scan itself.

[74] Dr. Hill suggested that while the 2008 MRI was not completely classical for perinatal asphyxia, it was indicative of it. The scan did show the involvement of the hippocampi, thalami and the dilatation of the temporal horns. Dr. Hill reminded the court that 20% of infants who experience HIE have normal MRIs.

[75] While Dr. Hill's evidence during his review of the two MRI scans was laboured and somewhat difficult to follow, and while he agreed that he would have to defer to a neuro-radiologist on some of the finer points, he was confident that both scans showed abnormalities which were germane to his opinion.

[76] Dr. Hill conceded that he could find no evidence of damage to the basal ganglia or the Rolandic cortex. He nevertheless clarified that any one or more of the involved areas of the brain that he referenced would be part of the acute near-total asphyxia pattern of injury.

[77] Dr. Hill was referred to a series of publications and papers which he had relied upon in formulating his opinion. Portions of these articles contradicted some aspects of his evidence in court. Nevertheless, many of those studies and articles also confirmed important points of his testimony. For example:

- A majority of children with moderate encephalopathy in the neonatal period go on to develop behavioural/cognitive problems;

- Approximately 77-90% of children with a history of neonatal encephalopathy and who suffer from cerebral palsy at 12 months of age are often also severely developmentally delayed;
- Whether the hypoxic-ischemic insult is the acute near-total type of injury or the watershed type injury, the majority of those afflicted suffered from cognitive and learning disabilities; and
- Such pregnancy and birth complications in mothers have been implicated in later diagnoses of ADHD in their children.

[78] I found Dr. Hill's evidence to be compelling and convincing. The overall quality of his discourse was informative, instructive and logical. He gave his evidence in a professorial fashion as if conducting a lecture in front of first year medical students; his vast curriculum vitae contained dozens of publications, reviews and presentations dealing with outcomes from perinatal asphyxia and HIE. I found his evidence to be highly reliable and entitled to great weight.

Dr. Robert Zimmerman

[79] Dr. Zimmerman was called by the Plaintiffs as an expert in Pediatric Neuroradiology. He has related expertise in perinatal asphyxia, HIE and brain injury. According to Dr. Zimmerman, the MRI scans confirmed that Sarah had sustained acute and profound asphyxia at birth.

[80] Dr. Zimmerman explained how the lack of oxygen flowing to Sarah's brain during the hypoxic-ischemic event caused damages to the thalami, hippocampi and central grey matter structures. Dr. Zimmerman pointed out the areas of damage on the imaging studies. The 2012 MRI was identical to the 2008 MRI thus confirming that Sarah's brain injury resulted from a single hypoxic-ischemic event. This event resulted in neonatal HIE, cerebral palsy and long-term developmental delay.

[81] Dr. Zimmerman was steadfast in cross-examination, adhering to his view that both the hippocampi and the thalami as imaged were abnormally small. While Dr. Zimmerman drifted outside of his core expertise in offering an opinion on outcome, I accept as genuine and credible his statement that he does have expertise in identifying the parts of the brain that are damaged and what such damage can mean for the patient. I accept the expert opinion of Dr. Zimmerman as reliable and place great weight on it.

Dr. David Berbrayer

[82] Dr. Berbrayer was qualified as an expert in Physical Medicine and Rehabilitation. He has wide clinical experience with cerebral palsy; he has written, presented and lectured extensively on the condition. He was called to give opinion evidence on both the diagnosis and prognosis for Sarah.

[83] Dr. Berbrayer conducted a functional assessment of Sarah at age 6; he reviewed her medical history before preparing his written report.

[84] On assessment, Sarah could count to 10 and sing the “ABC” song. She was not yet toilet trained. Sarah copied letters outside the lines and with shakiness. She demonstrated compromised fine motor skills. Sarah had a clumsy gait; she had challenges with stairs and on uneven ground. She had difficulty walking in a straight line; she had difficulty in turning.

[85] Sarah could not hop on either foot. She had floppy limbs. She was at risk of falling. She had difficulty articulating her speech. She drooled.

[86] Dr. Berbrayer identified the following entries in the medical records as worthy of note:

- Low Apgar scores of 4/10 at birth;
- Seizure activity requiring anti-convulsants after delivery;
- Intubation for 10 hours;
- A discharge diagnosis of perinatal asphyxia;
- Sarah was described as hypotonic;
- Sarah was referred to neurology at HSC;
- March 10, 2007 – Sarah remained hypotonic;
- July 9, 2007 – Sarah had motor development delay with no metabolic or genetic disorder;
- August 7, 2007 – Sarah had gross motor delay and truncal hypotonia;
- February 12, 2008 – Sarah was significantly behind in development. Her head circumference had dropped from the 10th percentile to 5th percentile. A referral was made for pre-school speech therapy;
- April 9, 2008 – development milestones progressing slowly;
- September 3, 2008 – diagnosis of hypotonic cerebral palsy;
- October 28, 2010 – safety issues, including clumsiness and stumbling, noted;
- January 25, 2011 – neurology clinic noted very slow progress;
- March 22, 2011 – psychological assessment showed attention and slow processing concerns;
- June 6, 2011 – Speech Language assessment revealed below age phonological awareness skills and a mild to moderate delay in speech production skills but normal receptive, expressive and social language skills;
- October 2011 – Occupational Therapy assessments;
- February 17, 2012 – JK conference identified weaknesses such as toileting, fine motor skills, speech articulation, focus and attention. Sarah was considered a “flight risk”; and
- April 16, 2012 – Dr. Logan conducted a pediatric neurologist assessment.

[87] Dr. Berbrayer’s diagnosis was perinatal asphyxia with hypotonic cerebral palsy as a result of a hypoxic ischemic episode at the time of the birth. Sarah’s condition was serious and permanent. Sarah had a reduced life expectancy due to increased risk factors associated with aging. From her age at the date of his assessment, she had a further life expectancy of between 60 and 65 years.

[88] Dr. Berbrayer also waded into the debate surrounding the academic performance of Sarah's siblings Isaac and Luke. He acknowledged that Isaac had some learning problems and that Luke had some initial difficulties but was now performing well. Luke had scored below Sarah on the Stanford-Binet Intelligence test prior to the twins' entry into school.

[89] In cross-examination, Dr. Berbrayer acknowledged that Dr. Logan did not characterize the staring episodes reported by the parents as seizures; however, Dr. Berbrayer maintained that what was described to him by Sarah's parents (infrequent episodes of staring lasting about one minute) would qualify as seizures. Dr. Berbrayer agreed that he would defer to the opinion of a pediatric neurologist on the issue of seizures. Dr. Berbrayer stated in his written report that Sarah had not had any further seizures since being discharged from hospital.

[90] Dr. Berbrayer made for an average witness. His testimony was generally consistent and reliable. I accept both his diagnosis and prognosis as accurate and reliable. His opinion that the staring episodes were seizures is undermined by the suggestion in his written report that the seizures did not continue after Sarah's release from hospital. None of the treatment providers, including Dr. Logan, would seem to support the notion that these staring episodes would qualify as seizures.

Dr. Phillip Wyatt

[91] Dr. Wyatt was called by the Plaintiffs as an expert in Genetics. He explained how the various genetic testing done on Sarah since birth had returned essentially normal results. Taking in to account the 2015 genetic testing, Dr. Wyatt concluded that there was no identifiable genetic component in Sarah's profile. In his view, the results of the genetic testing effectively eliminated the likelihood of a genetic component contributing to Sarah's deficits.

[92] Dr. Wyatt explained how technology in the realm of genetic testing continues to evolve. In this case, the microarray and exome sequencing conducted in 2015 revealed no alterations in any genetic material. The results were normal.

[93] Dr. Wyatt's opinion withstood a rigorous cross-examination in which he was obliged to delve into reports of the various treating doctors and their uncertainty about a definitive etiology. While Dr. Wyatt conceded that obtaining three generations of family history would be relevant for a geneticist, he was emphatic in his view that a genetic cause for Sarah's condition could be reasonably ruled out with the technology available in 2015. As well, he saw no justification to pursue testing of Sarah's parents that could run up charges of between \$50,000 and \$80,000 USD.

[94] Nothing came out of Defence counsel's exhaustive exploration of the difference between polygenic versus monogenic traits which caused him to waiver from his steadfast opinion. Dr. Wyatt's conclusions flowed logically from his analysis of the case study before him. I place great weight on his opinion and conclusions.

[95] Overall, I found Dr. Wyatt's expert testimony to be balanced, clear, informative, enlightening, educational and compelling.

Dr. Sharon Marcovitch

- [96] Dr. Marcovitch was called to testify by the Plaintiffs as an expert Psychologist. Dr. Marcovitch determined that Sarah suffered from a host of cognitive and academic deficits. She was of the view that the constellation of deficits and disabilities exhibited by Sarah were all related to the hypoxic brain injury suffered at birth. Sarah's behaviour was consistent with children afflicted by cerebral palsy and brain injury.
- [97] Dr. Marcovitch dismissed the suggestion by Dr. Jack Jung (a Defence expert) that Sarah's problems stem from inherited ADHD. Sarah's profile was distinct from that of her sibling Isaac, who had been diagnosed with ADHD. She added that a comparison of sibling behaviour at age four was not predictive of the future; a longer evaluation period was required. As well, there were notable differences between the two siblings on core skills. There was clear evidence that Sarah's condition stemmed from the injury at birth; there was no need to seek out other causal factors. To do so would be inappropriate and misleading.
- [98] Dr. Marcovitch concluded that Sarah's functioning difficulties were associated with neurological impairment stemming from hypoxic-ischemic brain injury rather than any inherited trait. A diagnosis of ADHD can only be made after ruling out other etiologies.
- [99] Dr. Marcovitch made recommendations for further assessment, investigation and treatment. She recommended speech-language therapy, occupational therapy, physiotherapy, an IEP, psychological assessment, certain assistive devices, computer programs, behaviour management, psycho-educational evaluation and accommodations at school.
- [100] Dr. Marcovitch found Sarah to have adaptive functioning levels well below her age in both receptive and expressive language skills and the understanding of words and letters. She noted significant deficits in memory, processing speed, motor skills and executive functioning. Sarah was delayed in her daily living, socialization and interpersonal skills. Sarah's motor skills were equivalent to those of children between 14 and 20 months of age.
- [101] Dr. Marcovitch made similar findings after a follow-up assessment in late 2015. Tests administered revealed that there had been little progress in Sarah's academic abilities; in fact, there was a widening gap between Sarah and her cohort. Sarah had a severe learning disability.
- [102] Based on these updated results, Dr. Marcovitch concluded that it was unlikely that Sarah would be able to enter into a college level program.
- [103] Dr. Marcovitch acknowledged Sarah's difficulties with attention/concentration, impulse control, hyperactivity and focus. Sarah faced challenges in executive functioning (i.e. organizing oneself). Sarah was found to be exhibiting anxiety, sadness and depression. There were problems with peer relationships, social withdrawal and aggression. Again, assessment findings in late 2015 were consistent with those made in 2013.
- [104] Sarah returned to Dr. Marcovitch for additional testing during the trial intake of various medications: Vynase in 2015-16 and Ritalin in 2016. These medications had made no impact

on Sarah's behavioural deficits. Dr. Marcovitch did not believe that medication would assist Sarah; rather, it might produce side effects of moodiness, headaches and sleeplessness.

[105] In cross-examination, Dr. Marcovitch conceded that she was not qualified to give an opinion on etiology. She agreed that in a psychological report authored by Dr. Ann Johnston in 2011, Sarah was reported as being extremely restless, impulsive and distractible. Dr. Johnston had suggested consideration and/exploration of ADHD if the situation did not improve with maturity. Dr. Marcovitch confirmed that Sarah was exhibiting significant levels of inattention, hyperactive/impulsive behaviour and impaired executive functioning.

[106] In her report of April 2013, Dr. Marcovitch had written that while Sarah met the criteria for a diagnosis of ADHD, her presentation and complex medical history were not typical of the condition. In that same report, Dr. Marcovitch supported recommendations and behavioural strategies to address and manage Sarah's difficulties with attention and self-regulation.

[107] Dr. Marcovitch was familiar with literature suggesting that ADHD was highly heritable. She acknowledged that two siblings could each suffer from ADHD with one exhibiting more severe symptoms than the other. While Luke's school records confirmed that he was struggling academically, he was not grappling with attention and impulsivity issues. Dr. Marcovitch was impressed with the level that Isaac had reached by Grade Seven; she doubted that Sarah would match that level of achievement.

[108] Dr. Marcovitch considered it only a small possibility that Sarah's ADHD was genetic; it was possible but not likely. Dr. Marcovitch noted that Sarah was inattentive and impulsive during the testing in 2013 and 2016. The growing gap in Sarah's abilities, confirmed by the Integrated Visual and Auditory Continuous Performance Test, was the result of the evolution of Sarah's neurological impairment and learning difficulties.

[109] Dr. Marcovitch effectively dropped the recommendations for medication, behaviour therapy and follow-up with a pediatrician contained in her 2013 report.

[110] I found Dr. Marcovitch to be credible; her evidence was detailed and comprehensive. While her evidence at trial about the possibility of ADHD appeared to be a retreat from the opinion she expressed in her 2013 report, she did qualify in that earlier report that given the complex medical history, Sarah's "presentation" was not typical of that condition. I see that caveat as critical.

[111] Although Dr. Marcovitch might have been more careful in her approach to the subject, I find the general tenant of her opinion persuasive: Sarah's profile includes ADHD-like symptoms but the diagnosis cannot be easily applied given the medical history. Indeed, what was germane to her opinion is that a diagnosis of ADHD cannot be made without ruling out other medical causes for Sarah's symptoms.

[112] I accept Dr. Marcovitch's opinion that Sarah continues to suffer from the deficits outlined. I accept that Sarah's academic skills and abilities are at a standstill and that the gap with her peers has widened. I accept that Sarah will not be able to match the achievement of her

siblings who are battling challenges of their own. I accept that she and her staff were able to work around Sarah's attention and distractibility problems to properly administer the most recent tests and obtain valid scores. I accept her evidence that medication has been ineffective in improving Sarah's deficits, including behaviour and attention problems.

[113] I accept Dr. Marcovitch's dim view for Sarah's post-secondary education; I accept her opinion that Sarah will not be competitively employable in the workforce. I also accept as reasonable her recommendations for various treatment modalities and assistive support moving forward.

[114] There were some problems with her evidence. Dr. Marcovitch is not qualified to ascribe Sarah's impairments to HIE. The abandonment of her previous support for behavioral therapy and medication was concerning. It struck me as an attempt to distance herself from any association with the Defendant's theory that behaviour therapy and medication were the answer to Sarah's deficits. That portion of her testimony should be looked at with skepticism; it does not, however, serve to diminish the reliability of her evidence which was, on the whole, thoughtful and convincing.

Dr. Mitchell Levine

[115] Dr. Levine was qualified as an expert in Clinical Pharmacology. His testimony was limited to the efficacy of medication on Sarah's behavioural performance.

[116] Dr. Levine focused on the two courses of medication: the trial of Vynase between December 2015 and January 2016, and the trial of Ritalin in May 2016.

[117] Dr. Levine based his opinion on the reporting by Dr. Marcovitch that there was no discernible benefit to Sarah during these trials. One would have expected to see an observable difference in a person within 30-60 minutes of taking this medication. This led Dr. Levine to conclude that it was unlikely that Sarah was going to respond to either of the two drugs in the future. He would have expected to see at least some response following the commencement of the dosage.

[118] In cross-examination, Dr. Levine acknowledged that neither the 20mg Vynase nor 5mg Ritalin dosages were ever increased. He never met Dr. Marcovitch, Sarah or her parents. He has no expertise in treating children with ADHD; he has no experience with pediatric neurology. He conceded that, depending upon the patient's weight, a common recommended dosage for Vynase could be 30mg. Medication doses are normally titrated. Sarah was never administered a non-stimulant medication; these are no more effective than stimulant medication in any event.

[119] I accept Dr. Levine's opinion. It was logical and persuasive. He had an excellent grasp of the subject matter of his expertise. His opinion was offered with clarity and conviction. Dr. Levine explained that without information about Sarah's weight, he could not conclude that the starting dosage of medication was insufficient; regardless, and notwithstanding that the dosage was not increased, one would have expected to see some immediate benefit from that initial dosage; we did not.

The Defendant's Evidence on Causation

Dr. Michael Shevell

[120] Dr. Shevell was called on behalf of the Defendant. He was qualified as an expert in Paediatric Neurology and neurologically based developmental disabilities. He has written extensively in the field of global developmental delay, ADHD and cerebral palsy.

[121] Dr. Shevell assessed Sarah in March 2015. Dr. Shevell acknowledged that Sarah had non-progressive cerebral palsy with a palsy-ataxic hypotonic variant.

[122] Dr. Shevell described the intrapartum asphyxic birth event as a near total one. He remarked on the neonatal encephalopathy and the acidotic cord PH level. Dr. Shevell also referenced the seizures, fetal heart rate changes and the Apgar scores in the neonatal period. Sarah's profile was mostly related to the intrapartum event.

[123] Nevertheless, Dr. Shevell then went on to make a diagnosis of both ADHD and Oppositional Defiance Disorder (ODD). Dr. Shevell based this on Sarah's clinical presentation (i.e. fidgety, restless and distracted) and the strong family history of learning disabilities (i.e. two of her siblings had learning disabilities).

[124] In his opinion, both the neonatal event and the ADHD were contributing to Sarah's condition and limitations. He was unable to assign any precise or relative contribution to the two conditions. Sarah's attentional limitations were hampering her ability to learn, to acquire new skills and to maximize her potential.

[125] Dr. Shevell believed that a properly administered trial of medication would bring about a positive benefit. The effectiveness of the 2015 trial of Vynase was compromised by both an insufficient dosage and ill-chosen timing.

[126] On the vocational front, Dr. Shevell noted that, while Sarah had some impairment in fine motor skills, her greatest challenges remained with learning, attention and behaviour. Dr. Shevell considered it premature to predict the outcome for Sarah; he was confident in forecasting, however, that with some degree of supervision, Sarah would be able to live independently in adulthood.

[127] In cross-examination, Dr. Shevell conceded that the hypoxic ischemic event caused injury to Sarah's cerebellum. The cerebellum is central to cognitive, social and language issues. The injury to the cerebellum in all likelihood caused Sarah's cerebral palsy. Although the hypoxic ischemic event probably contributed to her cognitive and behavioural issues, Sarah's family history was playing a part as well.

[128] Dr. Shevell was unable to locate any diagnosis of ODD in either sibling; he could not say if Luke had even been diagnosed with a learning disability.

[129] Dr. Shevell admitted to being the first medical doctor to make a diagnosis of ADHD; prior to his involvement, none of Sarah's treating doctors had made that diagnosis, recommended any treatment or prescribed any medication for the condition.

[130] Dr. Shevell did not list any of the Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition (DSM-IV) criteria for ADHD in his report. He conceded that hypoxic ischemic brain injury can cause symptoms that look like ADHD. He agreed that in making a clinical diagnosis of ADHD it is important to exclude other disorders that might overlap with it.

[131] Dr. Shevell agreed that Sarah has secondary microcephaly caused by brain injury. This has resulted in portions of her brain either having shrunk or having failed to mature normally.

[132] Dr. Shevell acknowledged that up to 32% of persons with cerebral palsy present with normal imaging studies; and that up to 60% of brain MRIs initially reported as being normal, were later reported to be abnormal when reviewed by expert neuro-radiologists.

[133] Dr. Shevell conceded that medication is less effective in children who have had a neurological injury; the side effects of medication for such children are usually more significant.

[134] I was not persuaded by Dr. Shevell's opinions and overall testimony for the following reasons:

- He was not a treating doctor; he could offer no reason why ADHD would not at least have been queried by any one of the treating doctors.
- He disregarded or ignored the proscription of a diagnosis of ADHD whenever an alternate syndrome, such as a brain injury, was present.
- He agreed with this statement from an ADHD clinical practice guideline: "to make a diagnosis of ADHD, the primary care clinician should determine that *Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition* criteria have been met (including documentation of impairment in more than 1 major setting); information should be obtained primarily from reports from parents or guardians, teachers, and other school and mental health clinicians involved in the child's care. The primary care clinician should also rule out any alternative cause." In spite of this guideline, Dr. Shevell did not list any of the DSM-IV criteria for ADHD in his report. He did not have the parents fill out any questionnaires; nor did he request or obtain any input or reports from teachers or service providers.
- While acknowledging the Canadian ADHD practice guidelines as a "good reference source", Dr. Shevell clearly did not follow any of the recommended procedures or protocol such as: an initial screening visit, a follow-up visit and use of a teacher assessment form. His justification for this neglect was baffling. Dr. Shevell undermined his own credibility before the court by suggesting that: physician adherence to the guidelines is irregular; adherence would take place in

an ideal world; few people have the time to fill out the suggested forms; and he would be surprised if people were following this guideline.

- Dr. Shevell agreed with the following passage from his report dated May 7, 2015: “Sarah merits five diagnoses as enumerated by myself. She clearly experienced an acute near total intrapartum asphyxial event, which can be attributed at a more probable than not level as the etiologic agent for her mild intellectual disability, ataxic hypotonic variant of cerebral palsy and her secondary microcephaly.” Yet, at trial, he attempted to excise the intellectual disability from that list. His attempt to retreat from what is clearly set out in his written report, by suggesting that “the etiologic agent” did not mean the exclusive cause, was transparent and, frankly, embarrassing. I find that he was clearly tailoring his opinion to conform with the Defendant’s theory of causation.
- I find that Dr. Shevell changed his opinion in his second written report when he added a family history of ADHD and ODD to the mix. Nowhere in his first report does he suggest that there was a family history of ADHD and ODD; he writes only of learning disabilities. When challenged on cross-examination about the discrepancy, Dr. Shevell could only offer, “I think I reflected on the case and elaborated on my original opinion.” Dr. Shevell’s opinion on this issue is greatly undermined by the fact that Dr. Blackstock, the psychologist who assessed Isaac at age nine, never diagnosed Isaac with ODD, even though Dr. Blackstock was able to assess Isaac directly in a clinical setting. As well, Dr. Shevell’s diagnosis of ODD must be viewed with some skepticism: he evaded the question posed in cross-examination about his clinical experience with ODD. Instead, he simply stated that, faced with behavioural issues which might be considered ODD, he acknowledges his own limitations and involves psychiatrists and psychologists to manage the issue. This is hardly the kind of clinical experience that would serve to undergird a reliable diagnosis of ODD.
- Dr. Shevell’s credibility was severely compromised during the exchange he had with Plaintiffs’ counsel about what he listed as supportive criteria for intrapartum asphyxia in his report dated April 2, 2015: the presence of neonatal encephalopathy of a moderate to severe variety, acidotic cord pH, fetal heart rate changes, depressed Apgar scores at five minutes, the need for case room resuscitation, the involvement of other organ systems (i.e. hematuria, hypocalcemia) as well as the EEG and imaging studies. For Dr. Shevell to state, “I gave a list of possible supportive features. Nowhere did I affirm, like I did with the essential features, that these were present in Sarah’s case” as a means of avoiding having to admit that the EEG and imaging studies were consistent with generalized encephalopathy, is simply mind boggling. This is especially so given that the features he listed are confirmed by other objective evidence to be the features present in Sarah’s case, including, in the case of the EEG, an entry in the RVH discharge summary that the EEG was consistent with generalized encephalopathy.

- Two crucially important concessions made by Dr. Shevell in cross-examination were that ADHD can form part of the sequelae of intrapartum asphyxia and that hypoxic-injury brain injury can cause symptoms that look like ADHD.
- The “strong family history of ADHD” that Dr. Shevell relies upon for his causation opinion amounts to nothing more than one diagnosis of ADHD in Isaac at age nine. This is thin to say the least. Indeed, Dr. Shevell was obliged to concede that many of Isaac’s character traits, exhibited behaviours and strengths were somewhat inconsistent with what one would expect to see in a child who has ADHD. Rather too conveniently, Dr. Shevell failed to consider: the parents’ condition; that Luke has never been formally or informally diagnosed with ADHD; and that sibling, James, is not only perfectly normal, but thriving.
- Despite the fact that he would have had the 2008 HSC microarray report available to him, Dr. Shevell chose to make no mention of it in his report. This was a glaring and unexplained omission of an important piece of evidence that would argue against any genetically inherited ADHD being part of Sarah’s profile.
- When faced with the genetic testing results from 2008 and 2015, Dr. Shevell stubbornly refused to concede that those results significantly reduce the possibility that the ADHD is genetically based. I find this refusal palpably irrational.
- Dr. Shevell inexplicably failed to mention the objective and succinct opinion offered by treating specialist Dr. Tam, who having reviewed the genetic and metabolic testing, concluded that Sarah’s difficulties and issues were attributable to hypoxic-ischemic brain injury.

[135] I find that Dr. Shevell’s dual diagnosis of ADHD and ODD is not reliable. It struck me as a poorly veiled attempt to put forward an alternative cause for Sarah’s present difficulties other than the brain damage caused by the hypoxic ischemic event.

[136] I reject Dr. Shevell’s opinion on causation and give little to no weight to his overall testimony.

Dr. Jo Anne Finegan

[137] Dr. Finegan was qualified as an expert in neurodevelopmental disabilities in children including ADHD, ODD and autism. She opined that Sarah had severe and untreated ADHD-combined type, ODD, impairment in executive functioning, neurocognitive weaknesses and learning disability.

[138] Dr. Finegan went on to suggest that Sarah had developed significant anxiety, an avoidant and disengaged style, and learned helplessness. Taken together with Sarah’s impulsive, ballistic response pattern, this precluded a valid and reliable assessment of her neurocognitive status.

[139] According to Dr. Finegan, given the family history of ADHD and learning disabilities, it is not supportable to attribute Sarah’s status to the birth events. Moreover, until Sarah is

treated, it is not possible to predict her educational/vocational possibilities nor assess a future care plan.

[140] Finally, Sarah has been damaged by the failure to implement treatment recommendations made in 2013. Had Sarah received this treatment, we would be in a better position to both characterize Sarah's residual strengths and weaknesses, and to make better informed recommendations and predictions.

[141] Dr. Finegan's overall evidence was entirely unconvincing for the following reasons:

- By her own admission, Dr. Finegan is a Clinical Child Psychologist. While she possesses an expertise in neurological disorders, she is not a medical doctor, pediatric neurologist, pediatrician or neuro-radiologist. Dr. Finegan cannot prescribe or administer medication. Dr. Finegan does not conduct genetic testing. Dr. Finegan does not provide care to neonates. Thus, for the purposes of causation, her opinion can be given only limited weight. Within the realm of HIE, cerebral palsy and genetics, she has no expertise or experience and must rely upon medical literature only. Indeed the ruling of this court was that she could not opine on the critical issue of "genetics v. ischemic event". Since the issue of causation largely boils down to the relative and respective contributions, if any, of HIE and ADHD to Sarah's present profile, I find Dr. Finegan poorly positioned to offer the court much assistance in determining that issue. I can also attach no weight to her opinion about the extent to which treatment would have helped to clarify the picture of Sarah we have before us. Much of the treatment for ADHD involves medication and since Dr. Finegan was not qualified to offer an opinion on the efficacy of medication, it follows that her opinion on how treatment would have clarified Sarah's profile can be afforded no weight.
- For much of her testimony in chief, Dr. Finegan simply read from her report dated July 8, 2015, rather than providing the court with a compelling and reliable *viva voce* presentation of her formulations. At times during examination in chief, she required cuing and redirection. Her oral testimony lacked conviction. In addition, I find Dr. Finegan to have been rather disingenuous in discussing her treatment recommendations for Sarah. In offering that "I would not want that for Sarah" when disagreeing with the suggested need for a full-time educational assistant and that "she [Sarah] is a real firecracker and she knows what she needs...I agree she knows what she needs" when considering the need for family psychological counselling, Dr. Finegan was looking to convey a genuine concern for Sarah's best interest. She spoke of the "best interests" of Sarah and her family. However, I am reminded that Dr. Finegan has not treated Sarah nor had any meaningful relationship with her. I find this proposed concern for the Butler family's best interest simply unworthy of belief as well as highly discrediting.
- Dr. Finegan's oral testimony was circumscribed by the court to the areas of expertise for which she was qualified. I find that, by widening the scope of her overview in the written report to include areas where she had no expertise but rather borrowed heavily from medical literature of qualified experts, she

sidestepped her duty to offer the court a fair and balanced opinion within the four corners of her expertise in favour of advocating the Defendant's position.

- I find that Dr. Finegan simply started with an assumed diagnosis of ADHD and worked backwards, parsing out those pieces of evidence that would support her opinion and discounting those pieces of evidence that would be inconsistent with it. For example, she glossed over a note in a recent school record from December 2015 which indicated behavioural improvement in Sarah.
- It was obvious that when administering an IQ test to Sarah, Dr. Finegan was selective in reporting the results. She admitted to leaving out low scores in perceptual reasoning, working memory and processing speed index. Dr. Finegan's explanation for doing so was that the omitted results were not reliable given Sarah's issues with distractibility and attention; this in spite of the fact that her notes from the interview reveal that, in fact, Sarah was motivated, focused and attentive at times. After some pressing, Dr. Finegan was forced to concede that Sarah would likely have scored in the 5th percentile had all of the results been tallied. The familiar refrain from Dr. Finegan was that Sarah's attention and behavioural problems rendered some of the test results unreliable. It is a strange coincidence that the test results that she did not factor into the final tally were those that would have driven down Sarah's intelligence score; and a lower score would serve to erode the theory that Sarah's current problems could be explained by untreated behavioural problems secondary to ADHD. As the trier of fact, I was disturbed by this poorly veiled attempt by an independent expert to help advance the Defendant's theory of the case.
- Dr. Finegan agreed that if a patient's symptoms are explained by another medical condition, one cannot make a separate diagnosis for ADHD. In her words, "if there is a current medical illness that has symptoms of inattention, no, you do not diagnose ADHD". Dr. Finegan offered the court no explanation as to why treating physicians, such as Drs. Poole, Hiram, Logan or Tam, made no mention of ADHD or why any of these practitioners would not have referred Sarah for an assessment or a trial of medication when each of them were well aware of Sarah's behaviour issues and familiar with her profile.
- It was only after a fuss that Dr. Finegan admitted that Luke had never been diagnosed with a learning disability and that in order to make such a diagnosis, a comprehensive DSM assessment would be required. As well, it was only with some discernible reluctance that Dr. Finegan acknowledged that a recent report card revealed how Luke, unlike his twin Sarah, was not exhibiting significant attention difficulties.
- Dr. Finegan admitted that it is not unusual for parents to hesitate in starting their young children on medication. She conceded that medication will not assist Sarah with her cognitive and academic problems.

[142] Overall, I reject the opinion of Dr. Finegan as biased, unbalanced and speculative. I would afford it no weight on the issue of causation.

Dr. Jack Jung

[143] Dr. Jung was qualified as an expert in Genetics. He testified that ADHD is a complicated polygenic trait that is highly heritable. Environment plays a smaller role in the causation of ADHD than does genetics. Because both intelligence and ADHD are polygenic, influenced by numerous genes, the testing done to date does not rule out a genetic cause for Sarah's cognitive and behavioural issues.

[144] Dr. Jung added that the whole sequencing test is not designed to look at polygenic traits or characteristics but rather for monogenic single gene causes of learning disabilities, intellectual disabilities, developmental delays and genome genes. Dr. Jung was not aware of any genetic, polygenic testing that could confirm a diagnosis of ADHD. The Baylor testing done on Sarah identified a number of "variants of unknown clinical significance in disease genes related to clinical phenotype".

[145] Dr. Jung's opinion failed to persuade for the following reasons:

- Dr. Jung refused to acknowledge the very obvious fact that none of Sarah's treating physicians had attributed her cognitive problems to a genetic cause until he was shown the consultation notes of Drs. Tam, Poole, Hiram and Logan.
- Dr. Jung's entire opinion was based upon testing done on Sarah and Luke at age four, and on Isaac at age nine. Dr. Jung agreed that, for differential purposes, assessments should be done over time. Indeed, this is consistent with Dr. Marcovitch's opinion which I accept as valid. In any event, even at age four, Isaac and Luke's overall cognitive and intellectual functioning scores were within the average range; whereas Sarah's scores were already below average.
- Monogenic factors had been ruled out. The only thing not ruled out were polygenic factors. It is not possible to check for these. There is therefore no concrete evidence that polygenic factors could even be *a* cause, let alone *the* cause, of Sarah's difficulties.
- Dr. Jung had no family history relative to either Sarah's parents or her sibling, James.
- Dr. Jung had no up-to-date information about Luke. In any event, Dr. Jung could not confirm that Luke had been diagnosed with an "intellectual disability".
- In preparing his report, Dr. Jung had no information about the findings of microcephaly in Sarah. Dr. Jung agreed that microcephaly in a child was significant and could be the result of brain atrophy.
- HIE can be responsible for attention deficits, hyperactivity and cognitive deficits; symptoms from that condition can mimic ADHD.
- If, as Dr. Jung suggests, the heritability of ADHD is 80%, then Luke or James, or both, should have ADHD. Neither of them do.

- Dr. Jung ended up conceding that Sarah's intellectual disability is likely due to her hypoxic ischemic birth injury and not to genetic factors.
- Dr. Jung was critical of the Baylor study because it did not cover all the genes that could have been tested; yet in the same breath, he advised that he was not aware of any genetic, polygenic testing that can confirm a diagnosis of ADHD.
- Dr. Jung's report did not address the likelihood of a person with ADHD having a sibling with ADHD. When asked about that in cross-examination, he appeared uncomfortable and slightly embarrassed before indicating that he would have to do research on that. While I am not being critical of Dr. Jung's inability to provide an answer to a question that he was not asked to address in his report, the court is left wondering why this rather straightforward query was not put to him from the outset.

[146] In the end, the opinion of Dr. Jung is simply not helpful to the court on the issue of causation because it does not provide a solid basis upon which one could conclude that genetics had any role to play in Sarah's current profile. The fact that a theory of causation "cannot be ruled out" does not raise it to the level of a probable or likely contributing cause. Raising a question does not prove a fact; nor can a mere suggestion that further testing might provide better answers serve as an evidentiary basis upon which to make a finding. Clinical findings of "variant of unknown clinical significance" mean nothing to the court unless some conclusion can be drawn from them. None can. Moreover, asking the Plaintiffs to demonstrate that a genetic cause for Sarah's deficits could be ruled out would be asking them to both prove a negative. At the end of day, what we do know is that the genetic testing that was done on Sarah showed no genetic abnormality. I prefer the opinion of Dr. Wyatt that the genetic testing that was done effectively eliminated the likelihood that Sarah's problems are genetically based.

Dr. Avrum Pollock

[147] Dr. Pollock was qualified as an expert in Paediatric Radiology and Neuroradiology. He offered an opinion on what could be gleaned from the four imaging studies done of Sarah's brain.

[148] He found no abnormalities on the January 26, 20007 ultrasound.

[149] The February 26, 2007 ultrasound revealed some fluid and prominence in the right cerebral hemisphere. The choroid plexi in the bilateral lateral ventricles were plump. He saw no scar tissue.

[150] The MRI scan from July 2008 demonstrated bilateral symmetric areas of increase T2 flare signal in the bilateral peri-atrial white matter. The brightness of that area compared to the surrounding tissue indicated more water at that location. That amount of brightness was more than one would expect to see in myelinated white matter. This is the last area of the brain to undergo myelination; however, this is a normal finding that one can see into adolescence. The hippocampal findings were non-specific; they might be seen on any patient of this age.

[151] In cross-examination, Dr. Pollock confirmed that Dr. Zimmerman was a renowned expert in both Paediatric Neurology and hypoxic ischemic brain injury. Dr. Pollock has authored no articles or publications in the area of hypoxic-ischemic brain injuries in newborns. Nor has he engaged in any studies or research into the subject. Dr. Pollock agreed that 20-30% of hypoxic-ischemic brain injured victims have normal imaging studies. Although he had not read the reports of either Dr. Shevell or Dr. Zimmerman, he disagreed that the imaging studies support the occurrence of a hypoxic-ischemic event. Dr. Pollock did agree that in an acute/near total hypoxic-ischemic event, any number of the structures of the brain can be injured; there is no pre-programmed order.

[152] I much prefer the evidence given by Dr. Zimmerman on the MRI scans. Dr. Zimmerman offered clear and instructive testimony during which he highlighted the various involved areas of Sarah's brain and provided the court with excellent insight into the vexing problem posed by the inconclusive imaging. As well, Dr. Zimmerman incorporated Sarah's medical history and the salient aspects of the birth event into his testimony. Dr. Pollock essentially ignored the clinical history in preparing his report and giving his testimony. Where the evidence given by Dr. Zimmerman conflicts with that given by Dr. Pollack, I prefer the evidence of Dr. Zimmerman.

[153] Dr. Pollock was all too anxious to justify his findings on the basis that they aligned with those of Dr. Shroff, who originally interpreted the 2008 MRI scan.

Dr. Manohar Shroff

[154] Dr. Shroff, the Paediatric Neuroradiologist who first reviewed the 2008 MRI, noted in his interpretative report the prominence of the temporal horns of the lateral ventricles with under rotation of the hippocampi and an increase in the signal of the white matter in the peritrigonal regions.

[155] Dr. Shroff confirmed that the 2008 MRI contained two major findings. He believed that the increased signal in the white matter could be indicative of brain injury. This is consistent with his statement on the interpretative report that there were no "other" significant abnormalities seen.

[156] While Dr. Shroff was not qualified as an expert, he explained that the area of the brain where he saw incomplete myelination back in 2008 would have been the last area of the brain to experience myelination between ages two and three. I find that his estimate of the time frame for myelination is useful because the MRI from May 31, 2012 is so similar to the MRI of 2008. Myelination should have occurred by the latter date (by which time Sarah was 5 years old). It had not. This supports a pathological process in the brain.

[157] Dr. Shroff also confirmed that obtaining clinical information, such as Apgar scores, blood gases, seizure behaviour and head circumference measurements, is crucial when interpreting MRI scans.

Conclusion on Causation

[158] The test for causation was set out by the Supreme Court of Canada (SCC) in *Clements (Litigation Guardian of) v. Clements*, 2012 SCC 32, [2012] 2 S.C.R. 181, at para. 8:

The plaintiff must show on a balance of probabilities that “but for” the defendant’s negligent act, the injury would not have occurred. Inherent in the phrase “but for” is the requirement that the defendant’s negligence was *necessary* to bring about the injury – in other words that the injury would not have occurred without the defendant’s negligence. This is a factual inquiry. If the plaintiff does not establish this on a balance of probabilities, having regard to all the evidence, her action against the defendant fails.

[159] The overwhelming evidence is that Sarah did suffer a severe hypoxic ischemic event at the time of birth. The constellation of signs and symptoms, including Apgar scores, blood gases, and seizures, in the immediate post-partum period are entirely consistent with this having been a serious event. To this must be added the consistency of the working diagnoses offered Sarah’s treating physicians:

- The RVH Paediatrician Dr. Vincent Ho;
- Developmental Paediatrician Dr. Hiram;
- The HSC Paediatric Neurologists: Dr. Logan and Dr. Duane MacGregor; and
- Family Physician, Dr. Poole.

[160] Indeed, the Defendant’s own geneticist Dr. Jung agreed that the hypoxic-ischemic brain injury can cause ADHD-like symptoms. Defence expert Dr. Shevell testified that the probable cause of Sarah’s intellectual disability was the hypoxic brain injury; and while offering to the court the possibility of genetic factors playing a role in her attention problems, Dr. Shevell confirmed that he did not, and could not, know if Sarah would have developed her current symptoms due to a genetic cause. It was simply beyond the realm of our knowledge.

[161] I have concluded that the entire genetic theory advanced by the Defendant must fail because it requires not only an unacceptable degree of speculation on the part of the trier of fact but also because it would require the Plaintiffs to prove a negative: that genetic information which might emanate from more sophisticated genetic testing would rule out a genetic cause for Sarah’s condition. I adopt the expert opinion of Dr. Wyatt, who impressed me as well qualified, knowledgeable, open-minded and enormously rational:

Sarah’s past and most current genetic testing were all normal, showing no identifiable genetic component contributing to her deficits with the technology we have available.

[162] Sarah was never diagnosed with ADHD by any treating physician; nor was any suggestion made that she be assessed for ADHD by any treatment provider or educator. I am not

prepared to place any weight on the isolated report of psychologist Dr. Johnston who, back in 2011, suggested exploring a consideration of possible ADHD in the future. It is apparent from her report that she did not even consider the impact of HIE on Sarah's profile. This report was copied to both Dr. Hiram and the neurology clinic at HSC; yet, no steps were taken by any of the treating doctors to explore whether Sarah had ADHD.

[163] Until the trial of medication in late 2015 and 2016, Sarah was never treated for ADHD despite the fact that her developmental delays were long standing and well documented. Indeed, her difficulties started from the day she was born. As admitted by Dr. Jung, the Defence geneticist, ADHD is usually first detected in a person only from the age of two onwards.

[164] Although she was not called to give evidence at trial, I place great weight on the view expressed by Dr. Tam in the consultation report dated November 13, 2014. I do so because this Paediatric Neurologist's conclusion was offered in the context of taking Sarah on as a patient some seven years after the birth event. By this point, Sarah's history was well documented. Having not been involved in Sarah's assessment or treatment until that point, I find that Dr. Tam was well positioned to offer a reliable opinion on the issue of causation. As well, her consideration of the question was untainted by the debate amongst the retained experts in the present litigation. The opinion is most persuasive because of its balanced but simple approach to the vexing problem caused by the inconclusive MRI results. In grappling with the question of whether Sarah's present symptoms are due to the hypoxic-ischemic brain injury suffered at the time of her birth, Dr. Tam came to a refreshing and entirely credible conclusion:

In summary, Sarah is a 7 year old girl with a history of prenatal (perinatal) asphyxia. I explained to the parents that although her brain imaging is not completely classic for prenatal asphyxia, it often isn't. The perinatal history does sound convincing for having had issues of hypoxic ischemia around the time of birth, and as she has not had any developmental regression or any other signs of any other etiologies this is likely the cause of her symptoms.

[165] At trial, Dr. Hill elaborated on Dr. Tam's simple but compelling commentary by explaining that there are many babies who, having suffered perinatal asphyxia, end up with cerebral palsy and other neurological problems but have normal MRI scans. Moreover, the medical literature on the condition confirms that a significant number of infants who have had hypoxic-ischemic brain insults do not have any brain imaging abnormalities.

[166] Taken at its highest, the Defendant's case on causation is that the probability of Sarah having ADHD is between 16% and 33%. If Sarah does in fact have ADHD, then the probability of it being inherited is less than 30%. On a simple balance of probabilities, the Defendant cannot establish statistically that Sarah has ADHD or that the ADHD-like symptoms she is exhibiting are an aspect of a genetically inherited condition. While the onus of proving causation does not rest on the Defendant, it strikes me as a highly complacent approach to the litigation process for a Defendant to raise a theory on causation and challenge the Plaintiff to disprove it.

[167] It is abundantly clear on the evidence that HIE can lead to symptoms that look like ADHD. It is an absolute certainty that Sarah suffered a hypoxic-ischemic brain injury. On the evidence before me, I am not satisfied that inherited or genetically based ADHD is a materially contributing cause of Sarah's cognitive, attention and behavioural problems. On the other hand, I find that, but for the birth event and the resultant HIE, Sarah would not have the host of attention, cognitive and behavioural problems which plague her.

[168] It is also plain and obvious on the evidence that the physical impairments and challenges facing Sarah are a feature of her ataxic cerebral palsy which is a direct result of the brain injury suffered during the hypoxic-ischemic event that took place on January 26, 2007.

[169] The balance of the evidence points to the hypoxic-ischemic brain injury suffered by Sarah at birth being the sole cause of her cerebral palsy, behavioural and cognitive deficits. I have carefully considered but dismissed the possibility of other etiologies playing a role in her present condition. On a balance of probabilities, I find that Sarah's past and present cognitive deficits, impairments and challenges result solely from the hypoxic-ischemic brain injury suffered on January 26, 2007. The Plaintiffs have proven causation.

Damages

[170] Sarah was 9 years and 4 months old at the commencement of the trial. She has been left with a host of physical, cognitive and psycho-social impairments. The preponderance of the evidence satisfies me that she now suffers from the following impairments, challenges and conditions:

- Ataxic-hypotonic cerebral palsy;
- Low muscle tone;
- Lack of coordination;
- Impaired fine motor skills with hand tremors;
- Impaired gross motor skills;
- Poor visual motor skills;
- Speech delays;
- Disturbed sleep pattern;
- Poor impulse control;
- Poor judgment;
- Cognitive deficits;
- Delayed adaptive functioning;
- Impaired social and emotional functioning;
- Intellectual deficits;

- Attention and behaviour difficulties;
- Drooling and speech impairment; and
- Poor safety awareness.

The Plaintiffs' Position on General Damages

[171] The Plaintiffs seek the maximum amount awardable by a court for general non-pecuniary damages. Based upon the report from expert Economist Professor Jack Carr, that amount would be \$363,200 expressed in 2016 dollars. The Plaintiffs cite a string of cases which stand for the general proposition that where injuries are catastrophic or devastating in terms of their impact on the plaintiff, the maximum general damages award is appropriate. It is the severity of the injury coupled with the ability of the injured person to appreciate her condition that should determine the quantum: *Dann (Litigation Guardian of) v. Chiavaro* (1996), 4 O.T.C. 331 (Ct. J. (Gen. Div.)), at pp. 66-67; and *Brito (Guardian ad litem of) v. Woolley*, 2001 BCSC 1178, at para 681.

[172] The Plaintiffs point to the case of *LaPlante (Guardian ad litem of) v. LaPlante*, 1993 CanLII 550 (BC SC), at para. 19, where a child aged six with a severe head injury leaving him with tremors affecting the right upper extremities and impaired mental capabilities with occasional behavioural problems was awarded the maximum for general damages.

The Defendant's Position on General Damages

[173] The Defendant's position is that Sarah's general damages should be assessed at between \$150,000 and \$200,000. Sarah has mild gross motor limitations. Her fine motor issues can be functionally addressed by occupational therapy. Sarah has a cognitive impairment but no expert can yet opine that she is incapable of post-secondary education or employment. The Defendant acknowledges Sarah's attention and social isolation issues but points to a failure to address her underlying behavioural issues; these impulsivity and attention problems can be expected to improve as she matures and/or is medicated.

Analysis – General Damages

[174] While I acknowledge that Sarah's injuries are serious and permanent and impact on many of her activities of daily living, it would in my view be an exaggeration of her present level of functioning to award maximum general damages. She can in no way be compared to the injured plaintiff in *Chow (Litigation guardian of) v. Wellesley Hospital*, [1999] O.J. No. 279 (Ct. J. (Gen. Div.)) who, after suffering hypoxic brain injury, was left blind, mute, a quadriplegic and with cerebral palsy; nor can her situation be compared with the plaintiff in *Crawford (Litigation guardian of) v. Penney*, 2003 CanLII 32636 (ON SC), who at age 19 was left requiring total care: he could only communicate through eye gaze, kicking with his left foot and a picture book. That plaintiff could not breathe in a sitting position, was constantly prone to respiratory infections, and could only be fed through a gastrostomy tube.

[175] Sarah is still able to enjoy a quality of life that would be denied to individuals with an all pervading series of impairments. She is able to: attend school, play and interact, enjoy pets

and television and operate a computer. She is not deprived of the dignity of independent function in many aspects of her life. She is not afflicted by crippling headaches or body pain. She is not confined to a wheelchair. She has not had to make use of a walker. She has been rated as a level two on the Gross Motor Function Classification System (GMFCS) which is consistent with her being able to walk independently while requiring some assistance going up and down stairs. Sarah's rating on the Manual Ability Classification System (MACS II), a measure of fine motor or manual abilities, was also a two; this equates with an ability to handle most objects but with somewhat reduced quality and/or speed of achievement. Surveillance evidence conducted by the Defendant shows Sarah wearing a back pack, running and climbing into the back of the school van. She walked into school without extraordinary supervision. She was able to twist her body to look behind her while walking forward, without stumbling or appearing unsteady. Further surveillance captured Sarah attempting a hand stand and rolling or flipping along the ground on her hands. While seen twirling at school, she did not fall, trip or appear to be unsteady. At one point, she was seen walking along some rocks or stones without falling.

[176] The term "pain" is certainly deserving of an expansive meaning when it comes to assessments of compensation; nonetheless, the term's plain meaning should also be respected. And while Sarah no doubt feels pain from tremors, social isolation, her own growing realization of the significance of her impairments and the frustration borne of cognitive deficits, there was no evidence that her everyday existence is plagued by any kind of relentless body pain, whole body incapacity or complete dependence. There is no independent confirmation in any medical or school records that Sarah continues to have bladder and bowel incontinence. While Ms. Butler may harbour on-going concerns in this regard, there is nothing corroborative of the condition which could lead me to include it as one of her permanent or ongoing conditions.

[177] No seizure activity has been confirmed since the neonatal period. Even if the staring episodes resemble a variety of seizure, there is simply no evidence that they are in any way disruptive, harmful, painful or frequent. Recall that Dr. Logan did not qualify them as seizures.

[178] To elevate every case of serious and permanent impairment to the level where it would qualify for maximum general damages recovery once it crossed a certain bar of "catastrophic" would have the dual effect of over compensating some individual claimants while diminishing the significance of the pain and suffering experienced by those who, on any sliding scale of compensation, would have undoubtedly recovered more for pain and suffering but for the cap imposed on the general damages by the Supreme Court in the *Andrews* trilogy: *Andrews v Grand and Toy Alberta Ltd.*, [1978] 2 S.C.R. 229.

[179] While I would agree that there is great room for improvement in Sarah's profile, I must also bear in mind: one, the grim forecast of Dr. Berbrayer that her situation is permanent; and two, whatever gains she does make will require her participation in hundreds of hours of therapy while others in her age cohort will be engaged in educational/ career pursuits and recreational activities, while generally advancing along a life trajectory that Sarah will never experience or enjoy.

[180] It is impossible to reconcile every case of pain and suffering with a comparable example. No two cases are identical. Assessments require individual crafting by the trier of fact after consideration of all of the evidence. The task in this case is made doubly difficult because the court never had the opportunity to meet Sarah or hear evidence from her.

[181] In *Cruz (Litigation Guardian of) v. Robins*, [2006] O.J. No. 3086 (Sup. Ct.), a 10-year-old plaintiff suffered injuries at birth leaving him with a learning disability, incontinence, sleep difficulty, balance problems, irresponsibility, and a brachial plexus injury. The court accepted expert evidence that the plaintiff was compromised somewhere between 30-40% and there was evidence that a pain syndrome was developing. General damages were fixed at \$250,000. The profile in *Cruz* is distinct from that seen here; nonetheless, I am persuaded that the damages here should be consistent with damages awarded in that generally similar case.

[182] Having considered all of the evidence, having accepted that Sarah suffers from the conditions, impairments, limitations and challenges listed above, and having accepted that her present condition is stable but permanent, I arrive at the sum of **\$250,000** as the amount to be assessed for pain, suffering and loss of enjoyment of life.

Awards Under the Family Law Act, R.S.O. 1990, c. F.3 (“FLA”)

[183] The Plaintiffs suggest that Mr. and Ms. Butler are each entitled to \$125,000 to compensate them for their respective losses of care, guidance and companionship that they could reasonably have expected to receive from Sarah but for the birth event.

[184] The Plaintiffs suggest that the siblings, Isaac and James, each receive the amount of \$50,000. The suggested amount for twin brother, Luke, is \$75,000. The Plaintiffs point to the unique relationship that one would ordinarily have enjoyed with a twin sibling.

[185] The Defendant argues that the amounts sought by the Plaintiffs are inordinately high. Damages of this sort should not serve to compensate for grief or the emotional impact that Sarah’s losses might have on the family: *Lord (Litigation Guardian of) v. Downer* (1999), 179 D.L.R. (4th) 430 (Ont. C.A.), at para. 12. FLA awards for non-pecuniary damages should reflect a measure of consistency with other like cases: *Matthews Estate v. Hamilton Civic Hospital*, [2008] O.J. No. 3972 (Sup. Ct.), at para. 205.

[186] The Defendant suggests that each parent is entitled to \$25,000. Siblings, Isaac and James, should be awarded \$10,000. Luke, Sarah’s twin, is entitled to \$15,000.

[187] I have concluded that the compensation under this particular head of damages must be confined to the loss of companionship since no evidence was led upon which any losses of care or guidance could be based.

[188] In assessing guidance, care and companionship damages, the particular family circumstances must be taken into account by the trier of fact: *Mason v. Peters* (1982), 39 O.R. (2d) 27 (C.A.), at pp. 28-29.

[189] The evidence shows that, without any real friends, social life or extra-curricular activities to speak of, Sarah has remained physically closer (and will remain so) to her mother than one what one might expect to see in an average mother-daughter relationship. The reality of course is that the companionship between them is clearly lacking in diversity, evolution and quality. While the Defendant suggests that Sarah helps her mother with cooking, baking and pulling weeds, and that Sarah will continue to share interests with her mother as she matures, this argument misses the fact that the quality of interaction will never be what it could have been between a healthy maturing daughter and her mother.

[190] While Robert Butler may not often be physically present with Sarah due to his work demands and his own past-times, one must acknowledge that any potential for a meaningful and rich intellectual relationship between the two is dead in the water because of Sarah's documented limitations.

[191] It strikes me as both challenging and daunting to adhere to case law that would call upon the court to consider particular family circumstances while at the same time attempt to achieve a measure of consistency with other cases. The reality is that no two cases are alike; family dynamics, because they are a product of interrelationships among persons whose lives, personalities, attitudes and outlooks change over time, are unique and difficult to compare.

[192] I agree with the Defendant that very little evidence was led in respect of the siblings. Still, Sarah is the only girl in a family of four siblings. She is technically the "baby" of the family. None of the children have yet left home. There is not a large age disparity among them. The siblings all still have their adult years ahead of them. Sarah's deficits are serious and permanent. One can only infer that the loss of "quality" companionship will be on-going and permanent.

[193] I have arrived at the conclusion that fair and reasonable awards for non-pecuniary damages under s. 61 of the FLA are as follows:

Recipient	Amount Awarded
Jaye Butler	\$50,000
Robert Butler	\$40,000
Luke Butler	\$25,000
Isaac Butler	\$20,000
James Butler	\$20,000
TOTAL	\$155,000

[194] I would award those amounts under this head of damages.

Damages for Future Losses

[195] The Defendant's position is that Sarah has only mild gross motor limitations. Her fine motor issues can be functionally addressed by occupational therapy. Sarah has a cognitive impairment but no expert can yet opine that she is incapable of post-secondary education or employment. The Defendant acknowledges Sarah's attention and social isolation issues but point to a failure to address her underlying behavioural issues; these impulsivity and attention problems can be expected to improve as she matures and/or is medicated.

[196] In *Andrews*, at p. 230, the Supreme Court laid down the principles upon which damages for future care should be based:

There is a duty to be reasonable. There cannot be "complete" or "perfect" compensation. An award must moderate and fair to both parties. Clearly, compensation must not be determined on the basis of sympathy, or compassion for the plight of the injured person. What is being sought is compensation, not retribution.

[197] In *Athey v. Leonati*, [1996] 3 S.C.R. 458, at para. 27, the Supreme Court elaborated on the approach to be taken on hypothetical or future events:

Hypothetical events (such as how the plaintiff's life would have proceeded without the tortious injury) or future events need not be proven on a balance of probabilities. Instead, they are simply given weight according to their relative likelihood....A future or hypothetical possibility will be taken into consideration as long as it is a real and substantial possibility and not mere speculation. [Citations Omitted.]

[198] It is settled law in Ontario that a plaintiff must show a real and substantial possibility or risk of a particular future pecuniary loss: *Graham v. Rourke* (1990), 75. O.R. (2d) 622 (C.A.), at para. 41.

Future Loss of Income

[199] The preponderance of evidence satisfies me that Sarah will not be competitively employable at any time in her life. I find compelling the statistical information offered by Dr. Perlman that 71% of those impacted by HIE had severe developmental delays and that many of those, who were not as badly off as Sarah, were unemployed or unemployable.

[200] As well, I attach great weight to the opinion of Dr. Marcovitch that it is highly unlikely that Sarah will be able to succeed at gainful employment. I find that the constellation of neurocognitive and physical deficits in Sarah is simply too great to allow for any kind of optimism on the vocational front; there has been no suggestion that any of these deficits can be reversed. Indeed, according to Dr. Berbrayer, these conditions are permanent. The real and substantial likelihood of Sarah remaining competitively unemployable was not seriously challenged by any defence expert, least of all Dr. Shevell who offered merely that Sarah's condition would no doubt pose vocational challenges for her in the future. I utterly reject the

all too convenient testimony of Dr. Finegan that we can only make sensible predictions about Sarah's vocational possibilities once she is treated.

[201] While there are many therapies that are reasonable and necessary for her condition, I find that these will serve only to maximize her potential for achievement in recreational, social, daily living and avocational endeavours. Sarah's adaptive and intellectual functioning are significantly delayed; her memory skills are extremely compromised. Sarah's academic skills are minimal; a large gap has formed between Sarah and her peers. Weak language and phonological skills compound the problems. Her attention and self-regulation limitations hamper her ability to learn. Those deficits will so limit her options for education that any academic endeavours undertaken will be applied and not designed for entry into the workforce. Indeed, as stated by Dr. Marcovitch, Sarah's deficits in executive functioning and her neurocognitive weaknesses will become more evident with time, creating further challenges to academic learning. I am satisfied that she will forever lack the skills, abilities, training and competence to engage in remunerative employment. I find that Sarah will remain competitively unemployable during her lifetime and that her earning capacity is effectively nil.

[202] Again, I do not accept that Sarah's progress has been or will be stymied by her failure to engage in behavioural therapy; I am merely satisfied that no degree of any treatment modality or therapy would have in the past or will in the future serve to alter the fundamental and overriding problems that have plagued her young life so as to make her a suitable candidate for the workforce. I accept that the proposed therapies, while reasonable and necessary to achieve optimal functioning, maintain an acceptable level of physical and neurocognitive functioning and provide Sarah with structure, goals and the opportunity for engagement, will prove, on the whole, insufficient to allow her to enter into the work force in any gainful or sustainable capacity. One has to look at the real world and consider her prospects in context. I am not persuaded that she will be able to make up the gap between her permanent deficits and what would constitute the lower end of performance expectations of an average arms-length employer.

[203] I accept that Sarah's pre-incident earning capacity (i.e. what she would have earned had she not suffered the neonatal injuries) would have been that of an average college graduate. This is based on Professor Carr's forecast that, given her own parents' education history, Sarah would have been able to attain a college diploma and enter into the work force at age 21. I find this to be a reasonable and probable forecast in light of the family's work ethic and work history. In particular, Robert Butler went on from high school to serve in the Canadian military. During his tenure there, he completed courses in explosives and radio operations. Sarah's siblings have shown the skill level and discipline to attain hunting licenses and scuba diving credentials. James is a straight A to B student.

[204] I find as well that Professor Carr took into account reliable productivity and employee fringe benefits data when projecting an income stream.

[205] It follows from my conclusion on causation that I can only reject Scenarios B and D offered by the Defence expert Professor Douglas Hyatt since they assign a 48% reduction in

earning capacity because of Sarah's pre-disposition to ADHD. I would also reject the approach of Professor Hyatt to include part-time or summer student work income as part of the average earnings for Ontario females aged 15-24. The inclusion of income for persons in this cohort would include full-time students working part-time to finance their education. It also is partly arrived at by considering only earnings from female high-school graduates, workers who were unemployed for most or some of the year and those who chose to work only part-time.

[206] I also find that by reducing employment earnings a further 13.8% for non-participation in the labour force, Professor Hyatt has effectively double counted the non-participation factor because he has already employed data that had been adjusted for that non-participation.

[207] Professor Hyatt also ignores income replacement possibilities when he considers the negative contingency of periods of disability.

[208] I prefer the approach of Professor Carr to fringe benefits which borrows from KPMG estimates for the value of employee fringe benefits such as pension plan contributions, extended health care plans and life insurance.

[209] I find that Professor Hyatt, in assuming a 100% probability of early retirement neglects to consider the positive contingency that Sarah might have worked past her assumed retirement age.

[210] I also find that Professor Hyatt fails to make any allowance for productivity beyond the first 15 years of the income loss scenario. In contrast, Professor Carr has relied upon historical trends to arrive at his conclusion that Sarah's earnings would have increased in real terms at a rate of 0.80% per year. After starting with direction found in r. 53.09 (of the *Rules of Civil Procedure*: R.R.O., 1990, Reg. 194) that the discount rate for periods beyond 15 years should be 2.5%, Professor Carr takes that productivity increase into account and adjusts the discount rate applicable to future income to 1.7%. I accept this approach. It is far more concordant with economic reality; as well, the court has the power to dispense with compliance with a rule of civil procedure where it is necessary to do so in the interest of justice. Since I accept Professor Carr's approach as soundly based on the history of average real wages and because I find that his calculation of income loss will more closely approximate the actual loss of income that Sarah will suffer, I find it in the interest of justice to dispense with strict compliance to r. 53.09.

[211] I am not prepared to reduce Professor Carr's future loss of income numbers by any general or specific contingencies because:

- I am satisfied that general contingencies, which represent "the common lot of all of us" (see: *Beldycki v. Jaipargas*, 2012 ONCA 537, at para. 79) are already implicitly contained in Professor Carr's reliance on average earnings of college educated females in Ontario; and

- There was an absence of compelling evidence that specific contingencies, ones that would be peculiar to Sarah, should apply. I am not prepared to speculate on mere possibilities.

[212] I conclude therefore that there is a real and substantial likelihood that Sarah will suffer a future loss of income in the amount of **\$1,881,846** expressed in May 2016 dollars and I would award that amount under that head of damage.

Past Extraordinary Care – Pecuniary Claim under s. 61(2) of the *Family Law Act*

[213] Sub-section 61(2) of the FLA provides for recovery of pecuniary loss for Sarah's family members as a result of her injuries. These claims may include nursing, housekeeping or other services provided. An award should be based on the value of services provided. The Plaintiffs' claim an amount of \$774,273.74 for these pecuniary damages to the date of trial. This will serve to compensate Mrs. Butler for the care she provided to Sarah that has been over and above what would have been expected from any parent in normal circumstances (referred to as "extraordinary care").

[214] The evidence establishes that Sarah's problems from birth have demanded an evolving set of supervisory and assistive type care with self-regulation, hygiene, dressing, toileting, prompting, reminders, extraordinary supervision and attendances at medical and therapy appointments.

[215] The Plaintiffs' rely on the methodology employed by Martha Binstock, Certified Life Care Planner, as a means by which an award for past extraordinary care can be made: two extraordinary hours per day starting from the first year of Sarah's life, gradually increasing to reflect the decrease in parental responsibility and involvement that one would expect to see in healthy and normally developing children. This methodology would see the hours of extraordinary care rise to:

- 4 hours per week in year 2,
- 6 hours per week in year 3,
- 8 hours per week in year 4,
- 10 hours per week in year 5,
- 12 hours per week in year 6, and
- 12 hours per week in years 7-10 (to May 16, 2016).

[216] The Defendant suggests that \$100,000 (before reduction for failure to mitigate) is more than sufficient to compensate Mrs. Butler for the extraordinary care provided to the date of trial. Mrs. Butler admitted that there was nothing special about the care Sarah required in the first year: the twins were both in diapers and required feedings. In year 2, the only extra care for Sarah was for physiotherapy exercises 3 times daily for 20 minutes plus some sporadic occupational therapy exercises.

[217] Half way through year 3, Sarah and Luke began to attend preschool. Sarah received all of her physiotherapy and occupational therapy at school. The only extraordinary care involved an occasional diaper change and providing water when Sarah woke in the night.

[218] Beginning in September 2011, Sarah was in full-time JK. She required some assistance with dressing. Infrequently, Mrs. Butler attended at Sarah's school to assist her with bowel movements. The after-school routine involved all of the children; nothing special was done for Sarah.

[219] For years 5 and up, Sarah was at school for 7 hours per day and asleep for 12 hours per day.

[220] The Defendant argues that up until age five, there was minimal extraordinary care. After age 5, the Defendant relies on Angela Fleming's (Occupational Therapist and Life Care Planner) assessment of 1,431 hours per year of extraordinary care (based upon 50% of total parenting hours for school and non-school days, 80 hours per year for toileting and 16 hours per year for medical appointments). At Personal Support Worker (PSW) rates, this amounts to \$22,638.42 per year or \$90,553.68 to the date of trial. Adding to this a modest amount for extraordinary care before age 5, the Defendant suggests that an award of \$100,000 under this head of damages is fair and appropriate.

[221] In *Dube (Litigation Guardian of) v. Penlon Ltd.*, [1994] O.J. No. 1720 (Ct. J. (Gen. Div.)), the trial judge was faced with mathematical calculations for determining the amount to be awarded for past extraordinary care. He preferred a simpler approach, stating at p. 78, "the sophistication of the calculations simply produces an illusion of reliability. I prefer to concede that the estimate of these losses is more art than science and proceed to the task on a simpler basis."

[222] In *Dube*, the trial judge approximated the value of services that were provided over certain specified periods of time.

[223] A similar approach was adopted by the trial judge in *Carere v. Cressman*, [2002] O.J. No. 1496 (Sup. Ct.).

[224] A different approach was taken by Spiegel J. in *Matthews Estate*. The trial judge determined that the use of the global approach would be inappropriate where there was ample evidence to support the amount of time spent and the nature of the services provided over and above what would ordinarily have been performed by the family member.

[225] I am persuaded that a precise mathematical approach is not possible in this case. Unlike in *Matthews Estate*, where all of services considered were necessitated solely as a result of the injury, we have in this case a blending of ordinary care with extraordinary care. It is difficult, if not impossible, to separate extraordinary care provided to children with disabilities from what would be ordinary care provided to perfectly healthy and normally developed children.

[226] While the literature relied upon by Ms. Binstock in arriving at her numbers cannot be ignored, nor can it be given the weight which the Plaintiffs suggest. Every case is different and depends upon so many variables. The various categories of children with disabilities are many. Ms. Binstock attempted to extract precise hours from past events. This was always going to be an impossible task because no two days were likely ever the same. In the absence of contemporary care records, time cards, diaries or calendars, her approach was doomed to be inaccurate and unreliable. This fundamental weakness in the methodology used by Ms. Binstock greatly diminished the quality and reliability of her calculations. This is not to be critical of Ms. Binstock or Ms. Butler: it simply persuades me that a precise or even near to precise retrospective calculation of the value of these past services is not possible.

[227] These problems were further accentuated by the concession made by Ms. Butler in cross-examination that up until the age two, there was little difference in the care provided to the respective twins. This admission undermines the basis for the Binstock opinion for years one and two. The Binstock opinion also pre-supposes a finding that Sarah required round the clock care even during sleeping hours and that Ms. Butler would not have required restorative sleep because of her other duties towards three other children. I found the evidence severely lacking in that regard.

[228] In approaching past extraordinary care, I remain mindful of the caveat from the Court of Appeal for Ontario in *Parsons Estate v. Guymer* (1998), 40 O.R. 445 (C.A.), at para. 21, that care also involves a caregiver being available to meet a person's needs. Nevertheless, given the tender ages of her children, the awesome responsibilities of parenting four children, including twins, the particular family dynamics (including that she was for some time before the birth of the twins a largely stay at home mother) and Mr. Butler's demanding work schedule, I find that Ms. Butler would have provided both a high level of supervision and an on-going full-time presence in the family home to her children even without Sarah's deficits and challenges. In my view, a person's "availability" must be analyzed in much the same way as other aspects of care; one needs to separate ordinary availability from extraordinary availability.

[229] I have no doubt that Ms. Butler was there for Sarah in many capacities from the time of her birth in January 2007. I accept that Mrs. Butler has been charged with duties above and beyond those of normal parenting. I accept that, as Sarah has grown, the gap between where she would have been but for her condition and where she was because of her condition, has grown. It follows that the amount of extraordinary care has increased over the years. However, there is little to no evidence of Mrs. Butler providing sophisticated Developmental Service Worker (DSW) level care to Sarah. There is little to no evidence that Mrs. Butler researched, studied or employed any DSW level care skills. The court received no evidence showing that Mrs. Butler worked with Sarah on her intellectual, cognitive or social functioning through the use of games, puzzles, worksheets, cue cards, activity sheets or computer programs. These items did not make their way into the evidence. It is clear from the evidence that Sarah received all of her occupational therapy and physiotherapy at school. In short, there is nothing upon which the court can assign a DSW rate to the majority of services provided by Ms. Butler to her daughter.

[230] In my view, the more appropriate rate to base compensation upon is that of a PSW. This rate would most closely approximate the value of the services provided to Sarah. Although there have been fluctuations in the market rate since 2007, and it appears that the rate is actually lower today than it was in 2013, I find it fair and reasonable to average out that rate at \$15 per hour for the time period between 2007 and the date of trial.

[231] I find that the estimates of hours advanced by the Plaintiffs and their experts for extraordinary care in the pre-trial period have been greatly exaggerated. Nevertheless, some calculations are necessary in order to arrive at a reasonable award for past extraordinary care provided to date.

[232] I find the number of extraordinary hours of care that will be required in the first few years of the post-trial period as estimated by Ms. Fleming to be reasonable. In light of the relatively flat line of Sarah's development in the last number of years, I find it reasonable to employ the post-trial estimates for extraordinary care to a retroactive assessment of extraordinary care provided to Sarah beginning at age four to the date of trial. Employing that methodology, I would calculate the value of past extraordinary care as follows:

Time Frame	Start Date	End Date	Number of Days	Hours Per Day	Hourly Rate	Annual Value of Extra Care
1 st year of life	February 14, 2007	February 14, 2008	365	1	\$15	\$5,475
2 nd year of life	February 15, 2008	February 15, 2009	366	2	\$15	\$10,980
3 rd year of life	February 16, 2009	February 16, 2010	365	3	\$15	\$16,425
4 th -9 th years of life	February 17, 2010	February 22, 2016	2191	4	\$15	\$131,460
10 th year of life	February 23, 2016	Trial Date (May 16, 2016)	83	4	\$15	\$4,980
TOTAL						\$169,260 (rounded to \$170,000)

[233] I would therefore award the sum of **\$170,000** under this head of damages.

Damages for Future Care

[234] In calculating damages for future care, the court must take into account both the injured person's best interests and the contingency that the future may differ from the experts' predictions at trial:

Damages for cost of future care are a matter of prediction. No one knows the future. Yet the rule that damages must be assessed once and for all at the time of the trial...requires courts to peer into the future and fix the damages for future care as best they can. In doing so, courts rely on the evidence as to what care is likely to be in the injured person's best interest. Then they calculate the present cost of providing that care and may make an adjustment for the contingency that the future may differ from what the evidence at trial indicates: *Krangle (Guardian ad litem of) v. Brisco*, 2002 SCC 9, [2002] 1 S.C.R. 205, at para. 21.

Life Expectancy

[235] I accept the evidence of Dr. Berbrayer that Sarah's life expectancy as of the trial date is approximately 69 years of age. Nothing was offered by either side to challenge that evidence. That assumption will undergird all calculations of future care costs.

[236] The parties agree that future pecuniary losses should be calculated from the date of the commencement of trial (i.e. May 16, 2016). I adopt the present value factors set out in Appendix VI of the FTI Consulting report dated April 5, 2016. The object of present value is to determine the value today of a stream of payments over a specific period of time in the future, which, assuming interest earned on investment, will pay out the sum of the payments perfectly over time. For the purposes of future care costs (as opposed to future loss of income), I also adopt the approach to discount rates as set out in the FTI Consulting report, which references r. 53.09.

Extraordinary Therapies

[237] I found the evidence offered by Katherine Van Gammeren, Andrea Switzer and Patti-Jill Roberts (the latter as it relates to occupational therapy services that are reasonable and necessary) compelling and reliable. Their testimony shared a common and compelling theme: given the host of permanent deficits being experienced by Sarah, speech-language therapy, physiotherapy and occupational therapy will be essential to ensure that the impact of Sarah's impairments are minimized, that her potential for improvement will be maximized and that she will have the tools, skills and support to meet the challenges brought about by changes in her life journey.

Speech-Language Therapy Services

[238] Ms. Van Gammeren, a Speech Language Pathologist, explained how Sarah's cerebral palsy has left her with impaired oral motor movements, decreased muscle tone, lack of coordination and weakness. Sarah has difficulties with both the direction and rhythm in her muscles. These impairments leave her with deficits in oral motor movements and planning. She has difficulties with repeating lengthy sentences, being understood, pronouncing the same word in a consistent fashion, breathing patterns and voice control.

[239] After conducting a two day interview and assessment at both Sarah's home and at her school, reviewing school and medical records and having Sarah complete some standardized and structured tests, Ms. Van Gammeren proposed that Sarah receive continued speech language therapy throughout her lifetime.

[240] The Defence expert Justine Hamilton found that Sarah demonstrated a host of speech-language therapy issues including: articulation difficulties, weak and uncoordinated speech, drooling, words searching, poor fine motor skills, weak phonological awareness and impaired saliva control. Assessment revealed that Sarah often had trouble understanding what others were saying and the meanings of words. Sarah scored in the 1st percentile on a Goldman-Fristoe Test of Articulation-Second Edition (GFTA-2) test designed to assess her pronunciation of certain words.

[241] Unfortunately, Ms. Hamilton adopted the highly complacent Defence theory of the case on damages in suggesting that speech-language therapy services would be of little utility unless and until Sarah managed to get her behavioural and attention problems under control. I find that this concern was entirely unsupported by any of the objective observers in Sarah's life. None of her treating doctors or teachers suggested this approach.

[242] I prefer the evidence of Ms. Van Gammeren. She provided concrete examples of the difficulties being exhibited by Sarah: sound substitutions, articulation substitutes, simplified blends, grammatical errors, inappropriate volume and close talking.

[243] Ms. Van Gammeren explained how Sarah's cognitive-linguistic and overall communication demands would change over her lifetime making speech-language therapy assessment and treatment necessary indefinitely. She offered the rationale that communication is an integral part of the activities of daily living. Individuals with cerebral palsy often have difficulty communicating effectively. Speech-language therapy is the best way to ensure that an individual's language skills are maximized.

[244] In breaking down the assessed needs, Ms. Van Gammeren identified the specific and evolving challenges that Sarah is expected to face during certain phases in her life. Ms. Van Gammeren maintained that speech-language therapy services will assist Sarah in adapting and coping with these changing circumstances. She explained how the frequency and cost of speech-language therapy would decrease over time, especially once Sarah's environment and skills acquired a level of constancy.

[245] In cross-examination, Ms. Van Gammeren was candid in conceding that Sarah's receptive language and cognitive linguistic skill were within normal limits and that sentence structure, word structure and expressive vocabulary were within normal limits.

[246] Ms. Van Gammeren disagreed with the suggestion that speech-language therapy services should wait until behavioral therapy was implemented. She added that behavioural problems are very commonplace in speech-language therapy cases. It is noteworthy that Ms. Van Gammeren's findings on assessment were generally consistent with those found in the Simcoe County District School Board Special Education Speech Assessment in January 2013.

[247] I find that there is a real and substantial likelihood that Sarah will require speech-language therapy services and assessments for the duration of her life. I find that there is a real and substantial likelihood that she will incur costs in line with the estimates of Ms. Van Gammeren. Given that the need for those services is directly linked to her cerebral palsy and cognitive impairments, Sarah is entitled to be compensated for the estimated costs of those goods and services up to her life expectancy of 69.

[248] The speech-language therapy services recommended, identified and quantified by Ms. Van Gammeren are reasonable and necessary to allow Sarah to maximize her communication and language skills. I would use her number, \$111,655, as the starting point for the calculation of the award. Given that Appendix IX of the CTI report dated April 5, 2016 uses the life

expectancy of 69 years, factors in present value and does not include services relative to an entry into the work force, the numbers set out in that appendix need not be reduced. I would, however, apply a 20% reduction on this amount for future contingencies, specifically the real and substantial possibility that Sarah will at some point experience no further benefit from speech-language therapy; determine that she can apply the learned skills without the need for formal speech-language therapy; and/or find that other therapies are providing her with adequate benefits without speech-language therapy. I would therefore apply the 20% contingency reduction to \$111,655 and award the sum of **\$89,325** (rounded) to the Plaintiffs for future speech-language therapy costs. I find that there is a real and substantial possibility that Sarah will incur those costs for speech-language therapy in the future.

Physiotherapy Services

- [249] Physiotherapist Andrea Switzer assessed Sarah in 2013. Her findings included: decreased tone, ataxia, fine and gross motor difficulties, unsteadiness, bilateral hand tremors and shaking, poor postural and truncal control, a clumsy and floppy gait. Sarah's low scores on standardized testing confirmed that her motor skills were significantly delayed. During the assessment, Ms. Switzer noted a lack of awareness for safety hazards as well as impulsive behaviour.
- [250] Ms. Switzer supported lifetime physiotherapy services, with a focus on the development of gross motor skills in the remaining childhood years. Further blocks of physiotherapy, including consultation, support and assessment sessions, would assist Sarah during her teenage years when body structure and weight changes will bring about new challenges.
- [251] The transition out of high school and into independent living would necessitate further physiotherapy intervention. Beyond age 20, the focus would be on maintaining acceptable levels of physical fitness and overall gross motor functioning.
- [252] Physiotherapy services and sessions should be complemented by orthotics which are designed to arrest any deformity and to promote proper balance and function.
- [253] The services should also be complemented by accessibility and mobility equipment which will assist Sarah in keeping up with her peers, improving efficiency of activity and overall function. Assistive devices are of special importance for persons with cerebral palsy as they experience a decline in their level of function over a number of years. A motorized scooter was recommended between ages 15 and 20; it was recommended that Sarah begin using a cane at age 40 and a walker at age 50.
- [254] The rationale for ongoing physiotherapy services into teenage and adult years is to assist the person to meet the challenges of changes in her environment: high school, the transition to independent living, as well as possible child bearing and child rearing responsibilities.
- [255] Physiotherapy support would remain important in the years between 40 and 50 because of the high risk of overuse injuries and decreasing mobility.

[256] As Sarah moves into her 50s and 60s, Ms. Switzer foresees physiotherapy as a means to manage the onset and consequences of osteoporosis.

[257] Between ages 60 and 70, the provision of a walker was thought necessary to help alleviate the risks associated with decreased mobility and balance issues. Continued use of orthotics was recommended for the increase in pain often experienced by persons in that age bracket.

[258] Ms. Switzer was not severely tested on cross-examination. I found her to be honest, forthright and consistent. When questioned about Sarah's impulsive behaviour during the assessment, she replied calmly that such patient behaviour is not at all uncommon in a clinical practice. Ms. Switzer pointed out the difference between occupational therapy (more concerned with fine motor skills) and physiotherapy (which focuses on physical skills). I accept her evidence that having both types of therapy would be important. Ms. Switzer defended the frequency of physiotherapy sessions in Sarah's childhood years and the need for consultations with her school on the basis that the earlier Sarah received the support, the better off she would be in the long run.

[259] Ms. Switzer did agree that by age 12, it is reasonable to expect Sarah will be physically able to manage self-care tasks. She pointed out how adults with cerebral palsy are more likely to encounter musculo-skeletal problems and loss of independent function than healthy functioning adults. Nevertheless, Ms. Switzer was prepared to concede that services for pregnancy were only a possibility and in any event would not likely be an annual cost during the age bracket of 30-40. As well, she agreed that the average person might require physiotherapy services between ages 60 and 70 for any number of ailments. I infer from the answer given that not all of Sarah's assessed needs in that age bracket could be considered extraordinary.

[260] In contrast to the evidence of Ms. Switzer, I found the evidence of her counterpart Mary Anne Lawton to be contrived and unreliable. Her testimony was palpably infected with the Defendant's theory on the futility of extraordinary therapies because of Sarah's behavioural issues. Ms. Lawton was eager, even excited, to tell the court how during her one hour assessment of Sarah, she noticed that Sarah was "extremely hyperactive" "moving continuously" "really quick" "constantly moving" and displaying "significant hyperactivity". Despite the fact that Ms. Lawton detected core instability, coordination and balance problems, poor postural control, pelvic weakness, awkward movement, hand weakness and tremors, she nevertheless advocated for depriving Sarah of significant and sustained physiotherapy services unless she was treated or medicated. She claimed that, without medication, Sarah's hyperactivity would interfere with her progress thus making physiotherapy services largely ineffective.

[261] In Ms. Lawton's view, 1-2 visits by a physiotherapist up until age 18, a total of 10-20 sessions between ages 20-50 and 4-5 physiotherapy sessions per year between ages 50 and 70 could be justified. As well, a smattering of horseback riding and other social physical activities would help improve Sarah's balance problems. On the other hand, medication would enhance Sarah's focus and attention.

[262] Ms. Lawton's evidence was unconvincing even before she was forced to concede the following points in cross-examination:

- Physiotherapy for a person suffering from cerebral palsy can help reduce the impact of impairments and improve that person's overall condition;
- Ms. Lawton saw Sarah for one hour in a clinical setting. Unlike Ms. Switzer, Ms. Lawton did not see and did not ask to see Sarah at home, at school or outdoors;
- Sarah has significantly reduced core and trunk stability;
- Sarah has intense tremors in the right hand;
- Persons with cerebral palsy may decline in gross motor function as they move into adolescence and young adulthood; and
- Within a coordinated comprehensive treatment plan, physiotherapy plays a role in both managing physical impairments and increasing mobility.

[263] When asked by the court to clarify her overall opinion, Ms. Lawton conceded reluctantly that persons with wide ranges of neurological deficits require and receive physiotherapy. If a treatment provider could manage her behavioural issues, Sarah was a good candidate for physiotherapy. In fact, Ms. Lawton conceded that Sarah had the "strong potential to be better than she is."

[264] I can afford no weight to much of the evidence of Ms. Lawton. I do not accept her suggestion that Sarah would not benefit from physiotherapy because of behavioural problems. I find her suggestion that Sarah be denied physiotherapy services that could greatly enhance her situation because of some behavioural problems to be both manufactured and untenable. I find that Ms. Lawton has greatly understated the amount of physiotherapy services that Sarah will require moving forward.

[265] I accept the greater part of the evidence of Ms. Switzer. I use her table of services as the starting point for the calculation of damages to be awarded for physiotherapy and accompanying services for Sarah.

[266] I deduct from that table any physiotherapy services beyond age 70. I further deduct from that table the assessment for possible job placement on the basis that it is unlikely that Sarah will be able to enter the work force in any remunerative capacity. I would further reduce the award for physiotherapy services related to child bearing and child rearing. I find that there is not a real and distinct possibility that Sarah will require services in this regard. I would therefore reduce the identified and quantified services by 20% to account for what would not be extraordinary expenses and for what are not reasonably likely to be required or incurred. I would therefore award the Plaintiffs damages for physiotherapy services and assessments in the amount of **\$169,938**, which is 80% of \$212,423.

Orthotic Assessments and Equipment

[267] In addition, I would allow an amount of **\$28,000** for the orthotic equipment (including assessments) at the rate of:

- 2 devices per year from 2016-2032;
- 1 device per year from 2032-2057; and
- 2 devices per year from 2057 to life expectancy as summarized in the orthotic equipment table prepared by Ms. Switzer.

[268] I accept the evidence of Ms. Switzer that orthotics in children and adults with cerebral palsy help prevent deformity and provide a base of support. I find that Sarah, rated as she is at GMFCS II, will begin to experience a decline in functioning in her late 40s or early 50s. This justifies the increase in orthotic assessments and devices in the last 20 years of her life.

Accessibility Equipment

[269] I am prepared to allow the costs of accessibility equipment (i.e. canes and walkers) at the present value amount of **\$ 3,745**. These items are listed in the accessibility equipment table prepared by Ms. Switzer. Not only will this equipment assist with Sarah's day to day activities and provide a safe and secure means of ambulation during difficult times and into her senior adult years, but it will also greatly alleviate the risk of her falling. I am not prepared to allow anything for a wheelchair as it is projected that she will not require such a device within her adjusted life expectancy of 69 years.

Occupational Therapy

[270] Patti-Jill Roberts was qualified to give evidence on Sarah's occupational therapy and equipment needs. She provided the court with a forecast of the goods and services that Sarah would require for the balance of her life expectancy. She also offered an opinion on Sarah's present and future extraordinary care needs.

[271] Ms. Roberts assessed Sarah at her home and at her school on May 23, 2013. Ms. Roberts familiarized herself with Sarah's personal and medical history. The assessment revealed that Sarah was not meeting developmental expectations. Deficits and impairments revealed on assessment included:

- Limited ability to independently bathe, dress and toilet;
- Impulsivity, behaviour and attention problems;
- Failure to manage the pace of her eating and a tendency to choke on food;
- Episodes of falling;
- Poor productivity at school;
- Tremors and spasms;
- Deficits in social/leisure pursuits; and

- Bowel and bladder incontinence.

[272] Ms. Roberts suggested that Sarah required 12 hours of daily extraordinary care. For 4 of those 12 hours, the care should be “skilled care” from either a DSW or a Rehabilitation Therapist. A skilled care provider would oversee the implementation of behavioural strategies and the reinforcement of safety behaviours. Ms. Roberts anticipated that this level and quality of extraordinary care would be required until age 18.

[273] Ms. Roberts anticipated that the bowel and incontinence issues would resolve by age 12. At that age, Sarah should be also able to physically manage self-care tasks in a timely manner; she would still require direct instructions for managing things like feminine hygiene products; and she would require verbal promptings to get through her morning routine.

[274] Ms. Roberts anticipated that Sarah will move out of her parents’ home at age 18; she would require one year of 24-hour care while adjusting to independent living.

[275] Between ages 19 and 65, the best case scenario would see Sarah learning and retaining some independent living skills and safety awareness. This would obviate the need for 24 hours of direct care. Extraordinary care needs would be limited to 18 hours of indirect/on-call services coupled with 6 hours of direct attendant care assistance from a skilled care provider; the latter services would provide Sarah with guided practice for independent living skills. By age 65, Sarah’s deficits will have been compounded by aging such that 18 hours of direct supervisory care would replace the indirect care services. Sarah would continue to require the 6 hours of direct attendant care from the skilled care provider.

[276] The worst case scenario would be if the anticipated improvements in Sarah’s independent living skills and safety awareness were not achieved; in that case, Sarah would require 24 hour direct supervision.

[277] Ms. Roberts made recommendations for occupational therapy treatment, equipment and assessments. In addition to an assessment at age 18 to assist in Sarah’s transition out of her parents’ home, she forecasted annual assessments for school and technology access to age 23.

[278] Ms. Roberts recommended weekly occupational therapy treatment sessions between ages 6 and 12 to address self-care, productivity, behaviour, safety, equipment use, internet safety and leisure activities.

[279] Occupational therapy sessions would convert to bi-weekly between ages 12 and 18. They would return to weekly again between ages 18 and 25 to ensure a smooth transition into adulthood.

[280] Ms. Roberts proposed a bank of 200 sessions to cover the adult years between 25 and 65. A further bank of 40 sessions would cover the post 65 period until the end of her life.

[281] Various therapy equipment was also recommended: weighted pencil grips, fidget toys, ergonomic positioning and various programs (such as Word Q, Kidspiration, and Kurzweil 3000). Most of these were one-time costs.

[282] Ms. Roberts reviewed Sarah's status on May 2, 2016. Sarah was then 9 years old. Sarah did not meet any of her developmental expectations. Poor core strength and posture persisted. She was not dressing independently; she had poor hygiene and grooming habits; a lack of safety awareness; she was unable to complete age appropriate chores. Sarah continued to stumble and fall up to three times per week. Sarah still had bowel incontinence at school. She could not clean herself after a bowel movement. Sarah still could not write her numbers. Sarah's gross and fine motor skills were well below expectation. Ms. Roberts concluded that the development gap between Sarah and her peers was widening.

[283] Safety concerns were still prominent: Sarah raced and stumbled down the stairs. She darted across the road. Ms. Roberts saw no real improvement in Sarah at age 9. Her previous recommendations for future care needs remain unchanged.

[284] In cross-examination, Ms. Roberts conceded that she did not confirm the reports of incontinence with Sarah's teachers; however, school reports had indicated that Sarah was not toilet trained.

[285] Ms. Roberts did not observe Sarah bathing, washing or brushing her teeth; the report of shovelling food came only from Mrs. Butler.

[286] Sarah was inattentive, impulsive and non-compliant.

[287] Ms. Roberts' recommendation for 24/7 supervision coverage was based upon Sarah's current level of functioning.

[288] Ms. Roberts agreed that the goal of occupational therapy is to improve activities of daily living and independence. Occupational therapy will help to improve Sarah's skills and functioning necessary for coping with permanent developmental challenges.

[289] Ms. Roberts confirmed that for 18½ hours of the school day, Sarah is either at school or asleep. She assumed that Sarah would move out of the family home at age 18 because the family is no longer legally obliged to provide her with care. By that age, all parental supervision is extraordinary. It is reasonable for Sarah to have access to an awake and alert caregiver who can provide care throughout the night.

[290] Ms. Roberts agreed that her recommendations total 120 sessions of speech-language therapy, occupational therapy and physiotherapy per year. Sarah has had no therapy outside of school since age two.

[291] Angela Fleming was called on behalf of the Defendant. Ms. Fleming was qualified as an expert in occupational therapy, rehabilitation and life care planning. Ms. Fleming provided two differing projections for Sarah's future care needs: one in which Sarah's ADHD was successfully treated and another in which it remained untreated.

[292] I found this approach to be of little assistance; I have found that Sarah does not have ADHD, but rather ADHD-like symptoms stemming directly from her HIE. These symptoms cannot be treated by medication. The 2015-2016 drug trial has proven this. All of the experts agree that medication will not assist a child with neurologically based attention behavioral deficits and impairments.

[293] During a robust and spirited cross-examination, Ms. Fleming conceded that Sarah was, at age 8 years and 4 months, functioning at a level equivalent to a child between the ages of 2 years, 6 months and 3 years, 1 month.

[294] Ms. Fleming agreed that Sarah required a high level of supervision.

[295] I would, however, accept Ms. Fleming's cautiously optimistic prediction on the prospective success of the various proposed treatment modalities:

[F]rom an OT [occupational therapy] perspective....I think that Sarah is going to be a very willing participant in intervention; I think she's going to be highly motivated. And I think the potential is definitely there that she could make these gains in all of these areas.

[296] The inherent problem with the Defence position is that in order for Sarah to make the gains that will ensure some acceptable level of independence and obviate the need for higher level supervisory care, an exceptional amount of therapy will be required. The Defendant suggests that the best evidence before the court is that Sarah will not experience functional decline due to her cerebral palsy until she is at least 50. I find, however, that in order to ensure that is not the case, or at least to minimize as much as possible the effects of that decline, on-going and consistent therapies will be an important feature of Sarah's routine.

[297] I do not accept Ms. Fleming's conclusion that Sarah's attentional issues impeded her ability to properly undergo formal testing or that the results reflect an underestimation of Sarah's skills. Ms. Fleming impressed me as rather too anxious to fall into locked step behind the Defendant's theory of the case.

[298] Ms. Fleming's testing found that Sarah presented with significant delays in all areas of functioning, including self-care and productivity. Questionnaires completed by Sarah's parents and teachers showed that Sarah's adaptive behaviour was well below normal for her age group. Why these results would be unreliable was never made entirely clear; for example, I find it odd that a handwriting sample could be affected by attentional issues. It would either be completed or not. Ms. Fleming never suggested that Sarah took too long to give the handwriting sample, refused to cooperate or walked away from the task. At the end of the day, Ms. Fleming was left criticizing the intensity and duration of occupational therapy services suggested by Ms. Roberts and justifying her conservative estimates for Sarah's occupational therapy needs because of concerns about therapy burnout and Sarah's current inability to participate, learn and apply the skills taught during therapeutic intervention.

[299] Even in her "untreated ADHD scenario", Ms. Fleming was more concerned with justifying limited occupational therapy for Sarah than with following her own assessment findings. I

find that Ms. Fleming's suggestion for the rate and intensity of treatment in either scenario falls well short of what Sarah will require in order to maximize her potential successes and minimize the prospects for an early decline in function.

[300] The Defendant has argued that the Plaintiffs' experts have recommended thousands of hours of useless therapy. I cannot agree with that suggestion because:

- The court received no expert opinion to that effect; and
- The criticism does not stand up to careful analysis. In my view, 134 hours of therapy per year is hardly excessive; it works out to just over 2½ hours per week. The Defendant then points to a total of "over 1000 OT [occupational therapy] sessions for Sarah over her lifetime." With a life expectancy of approximately 60 years from the date of trial, this works out to 16 occupational therapy sessions on average per year. Again, this seems hardly excessive, especially in light of how all the experts agree that occupational therapy will be of some benefit to Sarah.

[301] I have concluded that occupational therapy services will be reasonable and necessary to equip Sarah with skills and strategies for optimizing her level of function. Occupational therapy will assist Sarah in becoming the best that she can be. Occupational therapy is required on an on-going basis for the duration of Sarah's life expectancy.

[302] I am prepared to start with the full present value of the future occupational therapy costs, including both the technology access and community integration/safety assessments, as recommended by Ms. Roberts. This amount was \$177,500 (rounded). I would, however, discount that sum by 20% for future contingencies, specifically to allow for the real and substantial possibility that Sarah will experience periods in her life when she chooses not to engage in formal occupational therapy, is able to retain and apply learned skills without assistance or finds that other forms of therapy are providing her with adequate overall results. Applying that 20% contingency reduction to the \$177,500 results in a net figure of **\$142,000**. I find that there is a real and substantial possibility of Sarah incurring those costs for occupational therapy in her lifetime. I would therefore award that amount.

Behavioural Therapy

[303] Behavioural therapy of the kind recommended by Ms. Fleming is, in an ironic way, the bridge that links the two solitudes that are the respective positions of the parties. I have found that there is no basis upon which to deny Sarah other treatment modalities because of her attention and impulsivity problems; rather, I see behavioural therapy as part and parcel of the entire package of treatment.

[304] I find that behavioural therapy will assist Sarah with learning safety and independent living, self-regulation and self-care. It will also allow Sarah to achieve the maximum benefit from other forms of therapy. I heard no evidence that behavioural therapy would not be appropriate for a person exhibiting ADHD-like symptoms within a neurologically compromised profile.

[305] I find that Sarah will benefit from intensive and immediate short-term behavioural therapy of the kind recommended by Ms. Fleming in her Scenario 2 approach. It will perhaps be best left with the service provider whether that behavioural therapy should be short-term and intensive at age 10 or spread out over a longer period of Sarah's adolescent/teenage years. Regardless, I find that there is a real and substantial possibility that these behavioural therapy costs will be incurred by Sarah before age 18. I would award an amount of **\$34,000** (rounded) to offset the cost of essential behavioural therapy.

[306] I find that it is likely that Sarah will remain in her family home until age 25. At that time, it is reasonable to forecast that she will move into a supported independent living situation. Between ages 18 and 25, behavioural therapy should be available to help Sarah maintain and consolidate achievements reached during the intensive period of behavioural therapy in the adolescent and teenage years. Although behavioural therapy was not recommended *per se* by any of the experts for these years, it was part of Ms. Fleming's considerations when costing out private group home settings. It is clear that behavioural therapy would form part of the treatment regimen for Sarah if she were to enter a group home environment at 18. It follows that some degree of behavioural therapy is reasonable and necessary during the period after adolescence but before leaving her parents' residence. I would allow a bank of **\$35,000** for behavioural therapy during this time period.

Educational Supports

[307] I am not prepared to award an amount for the costs of an Educational Assistant (EA). Given that Sarah has been placed in a special education class at her school, I find it is more likely than not that EA support will be funded at the school level. Moreover, it is unclear to me whether a private EA would be permitted by the school board in light of union consideration. I am not prepared to find, therefore, that there is a real and substantial possibility that the Plaintiffs will incur EA expenses.

[308] I would allow for educational consultation on an annual basis to age 18. Strategies should be in place to help Sarah be compliant with school work. I find that this expense is likely to be incurred because of the critical shortage of school psychologists. This fact was admitted by Dr. Finegan. I would allow an amount of **\$15,000** as a one-time amount for this projected cost.

[309] I would allow for Central Auditory Processing Assessments at ages 10 and 12. This testing was recommended by both Ms. Binstock and Dr. Marcovitch. I am satisfied that, given Sarah's auditory working memory problems, this testing might shed some light on these issues and point to some interventions that might assist. I would allow **\$980** for this cost.

[310] The Plaintiffs did not lead any evidence about the need for or the cost of tutoring. It is not included in the tables of educational support costs filed as Exhibits 55 D and E. I am not prepared to allow anything for tutoring.

[311] Sarah will likely finish secondary school at age 18. I have determined that she is not a candidate for post-secondary education. I heard no opinion from a psycho-vocational or psycho-educational professional or expert that higher post-secondary education is feasible. It is apparent to me that Sarah will not be able to overcome the array of cognitive and intellectual deficits that continue to plague her so as to make her a viable candidate for college. I accept that she is functioning well below her peer level intellectually; that gap is expected to widen. She is best suited for day programming or other avocational pursuits. This will keep her engaged and stimulated and act as a supplement to the expected progress that she will make through the various therapies. This day programming will serve to fill the gap left in her life after high school but before she is ready and able to leave her family home in favour of supported independent living. Ms. Fleming costed day programming at \$17,810.00 annually. I would therefore allow for five years of day programming at a cost of **\$85,000**.

Additional Therapies

[312] I would allow for psycho-educational assessments. As well, given that Sarah is beginning to have some insight into the deficits and impairments that she has, I would allow for psychological counselling for Sarah and her family. In my view, any additional therapies that are put in place should be designed to maximize her achievements in activities of daily living, self-care, independence, quality of life and avocational pursuits. Efforts should be directed at setting Sarah up for success in these areas.

[313] I am prepared to allow for case management. The various therapies, equipment and assessments that await Sarah will undoubtedly pose a bewildering and intimidating array of new challenges and disruptions. Nevertheless, I am not prepared to allow for case management services on a full-time, on-going basis as suggested by the Plaintiffs for two reasons:

- One has to assume that the various therapists will be professional and dedicated enough to schedule their respective therapies in a co-ordinated fashion with other treatment providers and to keep Sarah's parents, teachers and health care providers abreast of all progress, setbacks and plans; and
- I find that case management will only be reasonable and necessary at the implementation and orientation stage of the various modalities.

[314] For these additional therapies and case management services, I am prepared to allow a one-time block amount of **\$50,000** to cover the costs of these services.

Avocational Activities

[315] Although recommendations were made for swimming lessons and horseback riding, I am not prepared to allow for these. I accept the evidence from Ms. Fleming that about 80% of school age children participate in extracurricular activities. There would be nothing extraordinary about these costs for Sarah.

Extraordinary Care and Supervision

[316] Ms. Binstock approached future extraordinary care costs on the basis that Sarah would require a measure of both basic supervisory care and higher level/skilled care. The average cost of a supervisory level caregiver was \$17.75. Rates for higher level care of the kind provided by Rehabilitation Therapists or DSWs averaged \$58.78 per hour. Ms. Binstock differentiated between rates charged for “live-in” care and “live-out” care. She calculated annual total care costs (including supervisory care, skilled care and respite for parents) to be \$117,105.09 from ages 7-18. From age 18 to Sarah’s life expectancy, care costs would rise to \$251,670.51 including an allowance for yearly finder’s fees.

[317] As outlined above, I am persuaded that Sarah will both require and benefit from speech-language therapy, occupational therapy, physiotherapy and behavioral therapy services throughout much of her adolescent, teen and early adult years. All of the therapists agree that the primary goal of these treatment modalities is improvement in every aspect of Sarah’s life. All agreed that these improvements could be expected. Surely it cannot be the case that dozens of sessions per year of this kind of high-end therapy will serve no purpose. Indeed, I find that that with proper treatment, including a good dose of behavioural therapy, Sarah will acquire and maintain the skills necessary to ensure her own safety and a gain good measure of independence in most of her activities of daily living. The latest school records in evidence (December 14, 2015) indicate an improvement in behaviour: she has stopped wandering; she is now asking for permission. These are major milestones. I am convinced that more will follow.

[318] Assuming as I have that Sarah will enlist or have enlisted for her the various therapies she requires, I see no reason why she will not acquire the knowledge, skills and adaptive techniques to meet many of the challenges that her life will bring. Indeed, the Plaintiffs’ occupational therapy expert agreed that it was reasonable to expect that Sarah would be physically able to manage self-care tasks by age 12. I am satisfied that this combination of therapies will help teach Sarah coping skill and strategies, wrong from right, danger from safety, how to react in a crisis well before the time that she is ready to leave the family home.

[319] As stated above, I find it most likely that she will leave her family home at age 25. I was not provided with any statistics on when persons with profiles similar to Sarah would typically leave the family home. Common sense tells me that it would tend to be at a later age, assuming that there is a supportive home environment. Indeed, that is the case here. I heard no evidence that Mr. Butler is close to retirement; that the family intends on moving; or that there is an understanding that Sarah will move out at age 18. Indeed, one is left to wonder why a claim for home renovations to accommodate Sarah is being brought at the same time that it is being forecast by the care cost assessors that she will leave the family home and attempt independent living approximately eight and half years post-trial. I am simply not persuaded that there is a real and substantial likelihood that Sarah will leave her family home and attempt independent living at age 18. That being said, I find that there is a real and substantial possibility that her transition to independent living will occur during her adult years between ages 18 and 25 and that the treatment modalities and assessments recommended will be necessary to prepare her for that.

[320] I do not accept the opinion offered by the Plaintiffs' experts that attendant care on a 24/7 basis will be reasonable or necessary at any time in the future. The evidence in support of this suggestion is thin. There appears to be little to no corroborating evidence from the school about incontinence problems. I am unable to find that Sarah is experiencing any seizure activities. I am not persuaded that Sarah is such a flight risk that she cannot be taught impulse control. I am not persuaded that she will remain at risk of falling and injuring herself as she grows into adolescence and young adulthood. Part of the many goals of physiotherapy will be to teach her balance, coordination and body control. I find it difficult to imagine that the provision of a stand up shower, a bathing bench and other appropriate safety features coupled with careful and repeated instruction on self-management from the therapists will not effectively eliminate the risk of her falling in the shower. There is no indication that Sarah will forget, fail to grasp or ignore the measures that she is taught and which are designed to safeguard her physical well-being.

[321] In my view, behavioural therapy would not only be beneficial for Sarah in teaching her basic self-regulation and safety, but it will also serve to obviate the need for direct supervision on such a sustained and unending basis. While Sarah is not going to be competitively employable owing to her cognitive, intellectual and physical conditions, this does not foredoom her to a life dependent upon attendant care and monitoring to safeguard her well-being. I find that there is no real or substantial possibility that Sarah will require the consistent high level of 24-hour supervision as proposed by the Plaintiffs' experts. Indeed, Ms. Butler agreed she did not want Sarah to live with a 24/7 caregiver if possible – she would sooner explore other options to allow Sarah to live as normal a life as possible.

[322] The combination of home accessibility renovations (see below), steady (even if delayed) progress in her learning at school, improvement in her adaptive skills, impulsivity and distractibility, attention problems, life skills and adaptive techniques all of which are not just goals but expected milestones justifying the plethora of treatment modalities and assessments, leave me satisfied that Sarah will not require ongoing supervision at the rate proposed by the Plaintiffs.

[323] I find that Ms. Roberts greatly exaggerated the attendant care required into the future. For example, her insistence that Sarah will require 12 hours a day of extraordinary care until age 18 is not tenable. On school days, Sarah is either asleep or at school for 18.5 hours, leaving only 5.5 hours in a 24 hour period. Moreover, Ms. Roberts contended that all of those 12 hours would constitute "extraordinary parenting". This is flawed reasoning; it allows for no deduction for the time that Sarah's parents would be providing parenting in any event.

[324] Ms. Binstock fell into the same error since her calculations make no deduction for ordinary care. As well, I find that both Ms. Binstock and Ms. Roberts failed to adequately factor in the opinions of most of the therapy experts that the recommended therapies would have positive effects. I find as a fact that Sarah's level of functioning, self-care and safety awareness will improve considerably with time, treatment and therapy, making the forecasts offered by Ms. Binstock and Ms. Roberts unreliable.

[325] I am much more attracted to the approach taken by Ms. Fleming on extraordinary care. I would adopt her suggestion for attendant care projected costs as much more realistic taking into account Sarah's overall profile and the anticipated improvements brought about by intensive, diverse and prolonged therapy and support. I agree with her that life care planners must assume the relative success of interventions. I agree as well that the suggested need for a skilled care provider for 6 hours per day is both unrealistic and excessive; the type of service from a high level worker is equivalent to therapy. Sarah will already be receiving an adequate amount of therapy. Finally, I agree that a supervisory caregiver (i.e. a PSW) is well able to provide structure and organization to Sarah without the need for a skilled care provider.

Ages 9-11

[326] I accept the evidence of Ms. Fleming that the intensity of the parenting required because of Sarah's overall condition was 50% above what would be expected from parents of perfectly healthy children. I found compelling her approach to add more hours of extraordinary parenting hours per year to account for medical appointments and for attendance at Sarah's school to address toileting issues. I accept as accurate that the annual extraordinary parenting hours for ages 9-11 would be 1,431. I find that it is both more realistic and more reasonable that the costs of the attendant care should be based on PSW rates. However, I prefer the evidence of Ms. Binstock who took an average from a cross-section of general care providers and arrived at a rate of \$17.75 per hour. This is a more realistic rate going forward; the \$15 for the past would have been an average rate for the nine years preceding trial. I therefore accept that the annual costs of extraordinary attendant care for Sarah commencing on May 16, 2016 would be \$25,400 (rounded). This equates to \$70 per day (rounded). Therefore, for the period running through to and including January 26, 2018 (Sarah's 11th birthday) I would award the amount of **\$43,540**.

Ages 12-18

[327] Ms. Fleming increased parenting hours per day by 0.5 hour, leaving extraordinary parenting hours at 2.75 daily. As in the years between ages 9 and 11, an allocation for medical appointments and extraordinary parent attendances at school was allowed, leaving total extraordinary parenting hours for this period in Sarah's life at 1,426 annually. Again, I find this reasonable and adequate to provide for Sarah's extraordinary care needs. Using the average general care provider rates referred to above, this amounts to an annual cost of \$25,311.50 for attendant care needs and a resultant total of **\$151,869** for that period in Sarah's life. I would award that amount.

Ages 18-19: Transition to Post-Secondary/Early Adult Life

[328] I would again adopt the rationale and approach of Ms. Fleming. Ms. Fleming supported the involvement of a DSW during the transition from secondary schooling to whatever post-secondary pursuits she might choose (which in my view, would likely consist of day programming). This would be a one-time cost at age 18-19 (the years 2025-2026). This type of intensive life skills/transitional support was also supported by Ms. Binstock. In my

view, it is reasonable to allow this intensive support for 40 hours per week at \$35 per hour over a 48 week period. This DSW support would also double as a higher end form of attendant care. This would be at a cost of **\$67,200**. I would award that sum.

Ages 19 to 25

[329] I find that Sarah will in all likelihood remain in the family home during these post-secondary/early adult years. I base this finding on the following:

- The parents' own wishes as expressed to Mrs. Butler that Sarah remain with them forever;
- The improbability that Sarah will study at the college level which might necessitate her leaving the family home;
- The improbability that Sarah will be competitively employable;
- The Butlers' expression that they have no intention of moving and no intention of retiring in the next 15 years; and
- Sarah's developmental delays will likely be a disincentive for her to seek independent living before age 25.

[330] I find that by age 18, Sarah's anticipated achievements, her learned adaptive skills, improved awareness, attention and focus on issues of safety and self-care, natural maturation, acquired familiarity with her home environment, the provision of safety equipment in the home, her use of mobility and orthotics devices and the improvements in function that one can expect from all of the treatment modalities will leave her in a position where she would not require intensive caregiving support moving forward. Nonetheless, given that her intellectual disabilities will persist, her situation would be best addressed, advanced and enriched by the provision of DSW services. I find that DSW services on an ongoing basis would be necessary to ensure that Sarah could successfully complete complex tasks such as budgeting, mail review and bill payments. DSW support will help to ensure that she continues to gain the maximum from participation in community-based programs and activities. As well, those DSW services would assist her in meeting challenges brought about by changes in both her life and the world around her (e.g. relationships, preparing to move residences, adjustments to personal losses, travelling, upgrades to account for changes in technology, obtaining a driver's licence). DSW services will assist Sarah in maintaining, recalling and implementing the adaptive skills, routines and safety measures that she learned during her adolescence and will continue to learn as the treatment modalities continue. These DSW services would also double as attendant care for any of Sarah's lingering supervisory care needs. I would award an amount of \$20,000 annually for these services for the six years between age 19 and age 25. This totals **\$120,000**.

Age 25-69: Supported Independent Living

[331] It is of course impossible to predict with any certainty what the future holds for Sarah. While wholly independent living is not appropriate for Sarah given her permanent deficits, it is in my view far more reasonable to forecast supported independent living (SIL). Not only

would this prove adequate to ensure Sarah's safety and well-being; it would also provide her with a measure of independence and with it, the dignity and positive self-image associated with living in a community. The ongoing therapies will ensure that Sarah has not only the physical skills to manage and contain her disability but also the learned behavioural skills to counter her tendencies exhibited at a young age to neglect her own safety.

[332] On all of the evidence, I am led to the conclusion that Sarah will, at or about age 25, move out of her family home seeking both a measure of independence and a type of accommodation that is suitable to her needs, her limitations and her life trajectory. I find that she will meet the qualification criteria given her host of documented permanent and serious deficits. I find that there is a reasonable and substantial likelihood that she will incur costs associated with supported independent living for most of her adult years.

[333] Ms. Fleming provided the court with some detail in respect of each category of SILs. She emphasized that for those accessing these services through private pay, there is no waiting list. Sarah would be able to choose any appropriate unit in any building. Ms. Fleming costed the following potential options:

- An Assisted Living Environment in an Ontario March of Dimes apartment. In this scenario, Sarah would have 24 hour access to care. As well, Ms. Fleming costed for direct purchase assistance of 2 hours per day of PSW support, 3 hours per week of DSW support and 15 hours per year of case management. The annual cost was found to be \$37,272.00;
- Developmental Services Ontario (DSO) Community Residences. This option would require Sarah to qualify as a person suffering from a developmental disability with onset prior to age 18. This option would feature Sarah living alongside 3-4 individuals with fully integrated 24-hour staff support and the shared services of a DSW. Sarah would have her own bedroom but there would be shared common areas. The annual cost would be between \$60,000.00 and \$85,000.00; and
- A Private Group Residential Setting which would have all of the features of DSO residences but with no need for any qualification. Again, for private pay, there would be no wait list. These private residential settings have day programming and behavioural therapy available. The annual cost at "Vita Community Living" (the example cited by Ms. Fleming) was \$105,021.00 including the behavioural therapy and day programming.

[334] I have concluded that Sarah would best be served by residing in a setting with three or four individuals each of whom would benefit from the availability of shared care. A March of Dimes apartment is not in her best interest: I heard no evidence that she would be involved in any daily socialization with others. In my view, it will be critical to have a consistent opportunity to socialize, interact and to feel part of a community. This is clearly in her best interest. She will receive that opportunity in options two and/or three as presented by Ms. Fleming. DSW support, behavioural therapy and day programming would also be available.

Sarah would be free to engage her own speech-language therapy, physiotherapy and occupational therapy services. Because the future is uncertain and because there is a risk of overfunding under this head of damage if I were to allow for the most expensive option, I find it reasonable and appropriate to take the average between the lower end costing of the DSO SIL and the higher end private residential setting SIL ($\$105,021 + \$60,000 \times \frac{1}{2} = \$82,500$ rounded). From that should be deducted the amount of \$15,000 for usual cost of living that Sarah would have incurred from age 25 onward had she not been injured. The net extraordinary average cost of \$67,500 for SILs accommodation is the result. Given that I have found that Sarah will begin to incur this extraordinary cost at age 25 (January 26, 2032), this will be an annual recurring cost for 44 years. Applying a present value factor of 24.41, the cost of the future supervised independent living is **\$1,635,500** (rounded). I would allow that amount.

Housekeeping and Home Maintenance

[335] I am not prepared to allow anything for housekeeping and home maintenance. I heard no evidence of what percentage of persons suffering from Sarah's litany of deficits and impairments would seek to purchase and maintain a home. I cannot find a real and substantial likelihood that Sarah will purchase or rent a home and go on to incur these expenses. I am also not prepared to infer that there would be any home maintenance expense in a supported independent living setting. Many of the areas in such settings are common areas. Given the cost of these residences per person, I would be surprised if some acceptable level of housekeeping services were not provided to the residents. In light of where I have forecast Sarah to be living beyond age 25, I find it unlikely that she will incur expenses for housekeeping and home maintenance in her lifetime.

Home Accessibility Costs

[336] Mr. Baum was qualified as an expert in accessibility housing for people with disabilities. He provided an opinion in respect of Sarah's needs and the projected costs of those needs in this regard.

[337] Mr. Baum attended the Butler family home in order to assess Sarah's extraordinary living requirements. He then prepared designs and costing for those recommendations. From the medical reports, Mr. Baum gleaned that Sarah had gait, balance and mobility problems, that she experienced difficulty on stairs and on uneven ground, and had fallen a number of times. He learned that Sarah was a flight risk. He learned that these deficits were permanent. Mr. Baum made the following recommendations for the Butler family home:

- A leveled paved driveway;
- An attached garage;
- A safe primary entrance with electronically controlled safety and lock devices;
- Bilateral handrails on staircases;

- A ramp, vertical platform or residential elevators for overcoming height differentials in the home;
- A safe secondary exit for emergencies and backyard access;
- A bedroom with adaptable closets;
- An accessible bathroom with vanity seat;
- A walk in shower with bench;
- A dedicated room for focus on tasks;
- Video and surveillance monitoring system;
- A fenced-in backyard; and
- Home modifications as shown on the architectural drawings.

[338] Mr. Baum explained that some of the proposed modifications required compliance with the *Building Code*: O. Reg. 332/12; as well, some of the changes to parts of the home necessarily involved alteration to adjoining parts. The proposed renovations would entail the elimination of a bedroom on the main floor; replacing it would require a small addition to the home.

[339] The total cost of the proposed accommodations including construction, disability-specific features and construction related expenses was \$279,400.00. There would be additional expenses: architectural consulting, project management, increased utilities, insurance and municipal taxes. There would be annual fees for elevator and lift maintenance.

[340] Mr. Baum anticipated that if the Butlers elected to proceed with the home renovations, they would likely continue to reside with Sarah at the present residence for the foreseeable future. Nevertheless he factored in an amount of \$60,000-\$85,000 for each time the Butlers moved residences to allow for core accessibility modifications.

[341] In cross-examination, Mr. Baum admitted that he took much of the information about Sarah's level of function from Dr. Berbrayer's report. He did not review any of the surveillance. Mr. Baum agreed that it was possible to create a covered access from the garage into the home without doing the internal renovations proposed: if the front stairs were enclosed and if Sarah was able to manage the existing stairs, then she could safely access the house. As well, the installation of railings at the back stairs would provide the secondary exit he envisioned. Similarly, the stairway to the basement could be closed in and railings installed to allow Sarah access to the basement.

[342] The existing bathroom would be able to accommodate a shower with a bench and a sitting vanity. Sarah could use the same room for both study and therapy. If Sarah could go up and down the stairs holding on to the railing, a lift in the backyard would not be necessary.

[343] The inclusion of the mudroom and displacement of the bathroom would not be necessary if the elevator was not installed. In that scenario, the master bedroom would stay in its current location.

[344] The Butlers advised him that Sarah would live with them forever. If Sarah were to move out at age 18 or 21, the future renovations discussed would be unnecessary.

[345] As stated above, I find that Sarah will continue to reside at her parents' home until at least 25 years of age. I am confident that the Butlers will go to the effort and expense of renovating the home to make it more appropriate for Sarah's safety, accessibility and therapy needs. However, major renovations of the kind suggested by Mrs. Butler are not necessary to address Sarah's safety needs: there is no evidence that she cannot climb stairs safely using hand rails. Again, the intensive physiotherapy which she will engage in will increase her core strength, and teach her adaptive techniques and skills for making her way safely around the house. Safety and flight concerns can be addressed with the video surveillance and locking devices as proposed. Given that Sarah was assigned a rating of GMFCS level 2 and that her subtype of cerebral palsy is ataxic hypotonic as opposed to the more debilitating spastic cerebral palsy, it is highly unlikely that Sarah will experience any kind of profound decline in her mobility functions before the age of 25.

[346] I would allow an amount for the video surveillance system with locking devices plus an amount for renovation of the home to include hand railings, enclosing the basement stairs, renovating the bathroom to provide for a stand-up shower with a bench and a sit down vanity.

[347] I would allow for the renovation of the home to include the study/therapy room for Sarah and the accompanying addition to create a new bedroom for Luke. Given Sarah's ongoing behaviour issues and the tenor of therapy that will be part of her everyday life, it would be in her best interest to have a study/therapy separated from the distractions of her bedroom and other parts of the home.

[348] I would also allow for the cost of a covered entrance from the garage and accessible front entrance.

[349] There should be an amount allowed for alternative accommodations for three months since the work will be disruptive and hazardous.

[350] There will be associated cost increases for utilities, insurance and municipal taxes. These will be modest given the limited scope of renovations allowed.

[351] I am not prepared to allow anything for vertical lifts or elevators. Sarah is perfectly capable of making use of the stairs in her home with the assistance of railings; I am not persuaded that there is a real and substantial possibility of her mobility declining so significantly in the next 15 years that she would require an elevator.

[352] Nor am I prepared to allow for the prospect of any residential moves by the Butlers in the next 15 years. It seems to me that once the necessary modifications are made to their home, there will be no incentive or purpose in a move.

[353] I am not prepared to allow an amount to Sarah for renovations to her residence of choice at age 25. In light of my finding that she will likely enter into a SIL, one must presume that

those future accommodations will be largely if not entirely suited to her needs. I heard no evidence that accommodations suitable to Sarah's profile would not be readily available in any number of SIL settings.

[354] Having considered all of her needs and circumstances, the likely course of events over the next 15 years, the gains that she will likely make from treatment modalities and her present level of functioning, I have concluded that the amount of **\$75,000** is a fair and reasonable amount for present and future home modifications. I find that there is a reasonable likelihood of such costs being incurred and would allow that amount accordingly.

Taxi Allowance

[355] I am not prepared to award damages for a taxi allowance. In the event that Sarah is eventually capable of taking driver training and obtaining a driver's licence, there is no real and substantial likelihood of there being any taxi expenses. On the other hand, if she is unfit to obtain or maintain a driver's licence, she will certainly have alternative transportation costs. These costs will, however, be entirely offset or minimized by the accompanying savings of not having to operate, finance, insure, service and maintain a vehicle.

Management Fee

[356] The Plaintiffs seek a 5% management fee on any award of extraordinary care costs. They argue that any amount awarded to Sarah should be invested in risk free securities in order to guarantee the availability of funds in the future as she acquires them. Sarah and her family do not have the requisite sophistication to do this; the cost to manage these funds would not have been incurred but for the injuries sustained; accordingly this represents a compensable loss to her.

[357] The Defendant argues that such a management fee is unnecessary because such a fee is likely to be self-financing.

[358] I agree with the Defendant. The proposed management fee is still subject to the real and substantial likelihood analysis. I can do no better than adopt the wisdom of my brother Quinn J. in *Dybongo-Rimando Estate v. Jackiewicz et al.*, [2001] O.J. No. 3826 (OSC) when he stated at para. 72: "A professional financial advisor should be able to obtain a rate of return (net of his or her fee) that is higher than...[the plaintiff]...would experience if he invested on his own, without professional help. Otherwise, how could such an advisor stay in business?"

[359] I am not prepared to allow for a management fee.

The Ministry of Health's Subrogated Claim

[360] Ontario Health Insurance Plan (OIHP's) subrogated claims were agreed to (subject to causation) at \$33,264.94 for past services and \$485.35 per annum for future services. As I have found causation in favour of the Plaintiffs, it follows that OHIP's subrogated claim for services rendered must be allowed for the total amount of **\$33,750.29**.

Government Funded Programs and Collateral Sources

[361] I am not prepared to reduce the award for future care, equipment and services on the basis that funding is now and may in the future be available from government agencies. As trial judge, I must make the awards for future care and services on the theory that the tortfeasors, and not any level of government, should be responsible for paying the cost of such items: *Lurtz v. Duchesne*, 2005 CanLII 2555 (ON CA), at para. 25.

[362] Sarah should not be forced to bear the risk of future reduced or eliminated government-funded assistance programs. If such programs were reduced or eliminated, or if the applicable access criteria were to change, Sarah might be left in the position where her damages award would be insufficient to fund those things that she requires to manage her disability. I adopt the wisdom of Dilks J. in *MacLean v. Wallace*, [1999] O.J. No. 3220 (Sup. Ct.), at para. 186, when he stated: “Funding under the Assistive Devices Programme is not mandatory; it depends upon the political and social conscience of the government of the day. In days of increasing cutbacks to social spending continuation of existing funding is by no means certain.”

[363] The same can be said to apply to virtually any government funded program. I am not prepared to expose Sarah to shortages of funding for her future care when those needs have been identified and quantified today and when she will have no right to come back to seek further compensation from the tortfeasor in the event that the identified programs are unavailable to offer future funding or assistance.

Reduction of Damages for Failure to Mitigate

[364] The Defendant argues that the Plaintiffs’ damages should be reduced by as much as 80% to reflect the probability that treatment for ADHD would have been successful if sought, provided and accepted.

[365] The court is obliged to assess the reasonableness of the Plaintiff’s actions or lack of action. In the case of *Labonte v. Sowers* (1994), 158 A.R. 350, at pp. 62-63, the court explained:

[T]he plaintiff must take reasonable steps to mitigate his loss. However, the courts are traditionally lenient in determining what actions are ‘reasonable’. The plaintiff need not act with perfect knowledge or ideal wisdom in moving to mitigate his loss. The plaintiff’s attempts at mitigation will be sufficient if they are reasonable and recovery will not be a bar because the defendant could suggest other remedial measures which are more effective. The determination of what is reasonable is always a question of fact.

[366] I am not persuaded that the Defendant has discharged its onus to prove failure to mitigate based on the following:

- I am unable to find on the evidence before me that Sarah even has ADHD. It remains at best a hypothetical and unreliable diagnosis which forms the

underpinning of the Defendant's position on causation. Statistically, there is no probability that Sarah has ADHD.

- The Butlers could not have sought to remedy a condition with which Sarah was not diagnosed. None of the treating doctors ever diagnosed, investigated for or suggested ADHD.
- It is apparent that Sarah has received some support at school for the attention and processing problems that have plagued her. Occupational therapy was re-introduced at school in December 2012. The Strength and Needs Committee has overseen her progress. Sarah has been in an IEP since 2012. This has provided her with special alternative programming for focus and attention. She has also benefitted from an educational assistant. In February 2015, at the prompting of an occupational therapist, Sarah was provided with strategies for classroom behaviour.
- The Plaintiffs did not act unreasonably in failing to obtain behavioral therapy for Sarah. The evidence has shown that a two year course of behavioral therapy would cost in the neighbourhood of \$34,000. It is profoundly unrealistic to expect a couple of modest means with four children living at home, to fork out that kind of money. I heard no evidence that such services would be covered under any extended health care plan. Had the Defendant thought that a course of behavioral therapy was the answer to Sarah's problems, it lay within its power to identify and quantify the cost of such therapy and to advance that sum to the Plaintiffs during the course of the litigation on a without prejudice basis. I heard no evidence that such an advance was offered.
- I am unable to conclude that the Plaintiffs somehow manipulated the failure of the medication trials. They administered Sarah the dosage prescribed by Dr. Poole.
- I am not at all persuaded that medication would bring about an improvement in Sarah's profile. I accept the expert evidence of pharmacologist Dr. Levine who testified that one would have expected to see even a small response from the dosage administered. The fact that there was no discernible change in Sarah led Dr. Levine to conclude that it was unlikely that Sarah was going to respond to a higher dosage of Vynase or Ritalin in the future. Moreover, although he did so reluctantly, Dr. Shevell was obliged to admit that up to 40% of children with brain damage will experience side effects from ADHD medication.

Reduction of Damages for the Contingency that Sarah would have ADHD in any event

[367] The Defendant argues that the Plaintiffs' pecuniary damages should be reduced to reflect the possibility that Sarah would have developed ADHD in any event. The Defendant suggests a mid-point range of 24% given that the evidence was that Sarah had a 16-32% chance of developing ADHD in any event.

[368] I am not persuaded that I should reduce damages as suggested by the Defendant because:

- There is no statistical probability that Sarah even has ADHD. The most sophisticated genetic testing available effectively ruled out the probability of a genetic component to her profile.
- There is no evidence, medical or otherwise, as to what Sarah's life would have looked like with ADHD but without HIE. Does ADHD prevent a person from pursuing an education, obtaining employment, living without assistance? There is no way for the court to know the answers without evidence. A person with genetically inherited ADHD can derive benefit from medication. But that is not the case here. The evidence is clear that a child like Sarah, having suffered HIE, can display ADHD like symptoms. It is just as clear that any medication prescribed for ADHD will be of no benefit to her; in fact, in 40% of cases, it will bring harrowing side effects.
- There is no way of assessing the notional damages that Sarah would have been entitled to had the HIE been superimposed upon her as a neo-natal with a pre-disposition to developing ADHD.
- I find that the dose administered by Dr. Poole was not inappropriately low. Dr. Shevell acknowledged that the recommended dose from the Canadian ADHD Practice Guidelines was between 20 and 30mgs. Those same guidelines recommended starting the child at the lowest possible dosage. Neither the Butlers nor the prescribing doctor acted unreasonably.
- As set out above, I find that Sarah's current profile, including her cognitive impairments and behavioural issues, are directly and causally related to HIE.

Conclusion and Disposition

[369] For the reasons set above, I would award damages to the Plaintiffs against the Defendant RVH as follows:

Category of Damages	Amount Awarded
General Damages Sarah Butler	\$250,000
<i>Family Law Act</i> Jaye Butler	\$50,000
<i>Family Law Act</i> Robert Butler	\$40,000
<i>Family Law Act</i> Luke Butler	\$25,000
<i>Family Law Act</i> Isaac Butler	\$20,000
<i>Family Law Act</i> James Butler	\$20,000

Future Loss of Income	\$1,881,846
Past Extraordinary Care	\$170,000
Future Speech Language Therapy	\$89,325
Future Physiotherapy	\$169,938
Orthotic Assessments and Equipment	\$28,000
Accessibility Equipment	\$3,745
Future Occupational Therapy	\$142,000
Behavioural Therapy	\$69,000
Educational Supports (including day programming)	\$100,980
Additional Therapies	\$50,000
Extraordinary Care: Ages 9-11	\$43,540
Extraordinary Care: Ages 12-18	\$151,869
Extraordinary Care: Ages 18-19	\$67,200
Extraordinary Care: Ages 19-25	\$120,000
Supported Independent Living: Ages 25-69	\$1,635,500
Home Accessibility Costs	\$75,000
OHIP Subrogated claim	\$33,750.29
TOTAL DAMAGES AWARD	\$5,236,693.29

[370] The Plaintiffs are entitled to judgment in the amount of \$5,236,693.29.

Pre-Judgment Interest, Costs, Periodic Payments and Ancillary Matters

[371] The awards for non-pecuniary damages and past pecuniary damages would attract pre-judgment interest at the rates prescribed in the *Courts of Justice Act*, R.S.O. 1990, c. C.43 (CJA). There remains as well the issue of costs. In the event that the parties are unable to agree upon the proper amount of interest on any award of damages and/or on the issue of costs, I invite them to make submissions on those issues in writing according to the following timetable: submissions of the Plaintiffs on or before May 31st, 2017; responding submissions of the Defendant on or before June 21st, 2017; and reply submissions, if any, of the Plaintiffs

on or before June 30th, 2017. The respective written submissions shall be limited to twenty pages, double spaced, with reply submissions limited to ten pages, double spaced. This page limit shall not apply to calculations of interest prepared by experts or to copies of Rule 49 offers to settle.

[372] I note that parties may consent to an order under s. 116 (1) of the CJA for periodic payment of all or part of the award for damages periodically. I note as well that since the award for future care exceeds the prescribed amount under s. 116.1(1) of the CJA that either the Plaintiffs or Defendant may make a motion that the damages for future care costs of the Plaintiffs be satisfied by way of periodic payments. There may be other ancillary issues arising from the judgment upon which the parties are unable to agree. If necessary, the parties may take out an appointment to address any remaining matters through the Barrie trial coordinator within 90 days of the release of this decision.

[373] I thank counsel on both sides for the excellent presentation of their respective cases.

J.R. McCarthy J.

Released: May 5, 2017